DISTRICT PANCHAYATH KASARAGOD

EQIP 2026

Educational Quality Improvement Programme for class -X

CHEMISTRY

ENGLISH MEDIUM

Standard -X

QUESTION POOL



DISTRICT INSTITUTE OF EDUCATION AND TRAINING (DIET)
KASARAGOD

EQIP 2025-26

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ആശംസ

വികേന്ദ്രീകൃത ആസൂത്രണത്തിലൂടെയും നിർവ്വഹണത്തിലൂടെയും കേരളത്തിലെ ആരോഗ്യവിദ്യാഭ്യാസ മേഖലകളെ ദേശീയ തലത്തിൽ ഒന്നാമതെത്തിക്കാൻ നമുക്ക് കഴിഞ്ഞിട്ടുണ്ട്. ഈ നേട്ടങ്ങൾ കൈവരിക്കാൻ പ്രാദേശിക ഭരണകൂടങ്ങൾ സ്തുത്യർ ഹമായ പങ്കുവഹിച്ചു. ദേശീയ സംസ്ഥാനതല പഠനങ്ങൾ നമ്മുടെ കുട്ടികളുടെ പഠന നിലവാരം ഇനിയും ഉയരേണ്ടതുണ്ട് എന്ന സൂചനയാണ് നൽകുന്നത്. പത്താം ക്ലാസിലെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഹരിക്കാൻ കഴിഞ്ഞ കുറെ വർഷങ്ങളായി കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ പൊതുവിദ്യാഭ്യാസ വകുപ്പും കാസറഗോഡ് ഡയറ്റും ഒത്തു ചേർന്ന് നടപ്പാക്കുന്ന **'എക്വിപ്പ്' 2025-26 (EQIP 2025-26)** പഠനപരിപോഷണ പരിപാടിക്ക് എല്ലാ പിന്തുണയും ഉറപ്പുതരുന്നു. പൊതുപരീക്ഷകളെ അഭിമുഖീകരിക്കുന്ന പത്താം ക്ലാസിലെ കുട്ടികളുടെ പഠനപ്രവർത്തനങ്ങളുടെ മികവിന്റെ അടയാളമായി മാറുകയാണ് വാർഷിക പരീക്ഷകൾ. അറിവിന്റെ തെളിമയോടെ ഓരോ വിദ്യാർത്ഥിക്കും പരീക്ഷ എഴുതാൻ കഴിയണം. വിദ്യാർത്ഥികളുടെ ജീവിതത്തിലെ ഏറ്റവും പ്രധാനപ്പെട്ട പരീക്ഷകൾക്ക് വേണ്ടി തയ്യാറാക്കിയ ചോദ്യബാങ്കിന് എല്ലാവിധ ആശംസകളും നേരുന്നു. നന്നായി പഠിക്കുക. പരീക്ഷയെ സധൈര്യം നേരിടുക. തളരാതെ മുന്നോട്ട്. വിജയം നിങ്ങളോടൊപ്പമുണ്ട്. ആശംസകൾ.

ശ്രീമതി ബേബി ബാലകൃഷ്ണൻ
ജില്ലാ പഞ്ചായത്ത് പ്രസിഡന്റ്
കാസറഗോഡ്



ആശംസ

കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ജില്ലയിലെ പൊതുവിദ്യാഭ്യാസ മേഖലയെ ശക്തിപ്പെടുത്തുന്നതിന് നിരവധി പ്രവർത്തനങ്ങളാണ് നടന്നുവരുന്നത്. പൊതുവിദ്യാഭ്യാസ മേഖലയെ പൂർവ്വാധികം കരുത്തോടെ നാം മുന്നോട്ട് നയിക്കുകയാണ്. ഈ ഘട്ടത്തിലാണ് കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തും, പൊതുവിദ്യാഭ്യാസ വകുപ്പും, വിദ്യാഭ്യാസ പരിശീലന കേന്ദ്രവും **(DIET)** പത്താം ക്ലാസ് വിദ്യാർത്ഥികളുടെ പഠന വിടവുകൾ പരിഹരിക്കുന്നതിനും ആത്മവിശ്വാസത്തോടെ പൊതുപരീക്ഷയെ നേരിടാൻ അവരെ പ്രാപ്തരാക്കുന്നതിനും വേണ്ടി പഠനപരിപോഷണ സാമഗ്രി തയ്യാറാക്കിയിട്ടുള്ളത്. ഈ പദ്ധതിയുടെ ഭാഗമായി ആത്യന്തിക വിലയിരുത്തലിനെ അഭിമുഖീകരിക്കാൻ കുട്ടികളെ സഹായിക്കുന്ന തരത്തിൽ വിവിധ വിഷയങ്ങളിൽ വിലയിരുത്തൽ സമീപനത്തിനനുസരിച്ചുള്ള ചോദ്യശേഖരമാണ് **(Question Pool)** ഈ അക്കാദമിക വർഷം തയ്യാറാക്കിയിട്ടുള്ളത്. കുട്ടികളുടെ അക്കാദമിക് മികവ് ഉറപ്പുവരുത്തിക്കൊണ്ട് മികച്ച ഗ്രേഡുകൾ നേടാൻ അവരെ സജ്ജമാക്കാൻ **എക്വിപ്-2025-26 (EQIP-2025-26)** എന്ന പേരിൽ തയ്യാറാക്കിയ ഈ പദ്ധതിക്ക് കഴിയട്ടെയെന്ന് ആശംസിക്കുന്നു.

സ്നേഹപൂർവ്വം

അഡ്വ. സരിത എസ്.എൻ.

ആരോഗ്യ-വിദ്യാഭ്യാസ സ്ഥിരം സമിതി അധ്യക്ഷ,

ജില്ലാ പഞ്ചായത്ത്, കാസറഗോഡ്



ആശംസ

ജില്ലയിലെ അക്കാദമിക പ്രവർത്തനങ്ങളെ ഏകോപിപ്പിച്ച് മുന്നോട്ട് നയിക്കുന്നതിന്റെ പ്രധാന ഉത്തരവാദിത്തം കാസറഗോഡ് ജില്ലാ വിദ്യാഭ്യാസ പരിശീലന കേന്ദ്രത്തിന്റെ (DIET) നേതൃത്വത്തിൽ കാലങ്ങളായി ചെയ്തുവരുന്നുണ്ട്. പ്രീ-പ്രൈമറി തലം മുതൽ ഹയർസെക്കണ്ടറി തലംവരെയുള്ള മേഖലകളിൽ വിവിധങ്ങളായ പദ്ധതികൾ ആസൂത്രണം ചെയ്യാനും നിർവഹിക്കാനും ഡയറ്റിന് ഇതിനകം കഴിഞ്ഞിട്ടുണ്ട്. പത്താം ക്ലാസ്സിലെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഹരിക്കാൻ കഴിഞ്ഞ കുറച്ചു വർഷങ്ങളായി ഡയറ്റിന്റെ നേതൃത്വത്തിൽ വിദ്യാഭ്യാസ വകുപ്പ് നടപ്പിലാക്കുന്ന പദ്ധതിയാണ് **എക്വിപ് (Educational Quality Improvement Programme).** അതാതു വർഷത്തെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഗണിച്ചുകൊണ്ടാണ് പ്രവർത്തനങ്ങൾ ആസൂത്രണം ചെയ്യുന്നത്. പത്താംതരം പാഠപുസ്തകവും വിലയിരുത്തൽ രീതിയും മാറിയ ഘട്ടത്തിൽ ആണല്ലോ നമ്മൾ. അതുകൊണ്ട് ഈ പദ്ധതിയുടെ ഭാഗമായി പത്താംതരത്തിലെ ആത്യന്തിക വിലയിരുത്തലിനെ അഭിമുഖീകരിക്കാൻ കുട്ടികളെ സഹായിക്കുന്ന തരത്തിൽ വിവിധ വിഷയങ്ങളിൽ പുതിയ വിലയിരുത്തൽ സമീപനത്തിനനുസരിച്ച് ചോദ്യശേഖരം (Question Pool) പരിചയപ്പെടുത്താനാണ് ഇപ്രാവശ്യം ഉദ്ദേശിക്കുന്നത്. ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ഡയറ്റിന്റെ അക്കാദമിക സഹായത്തോടെ മലയാളത്തിലും ഇംഗ്ലീഷിലും കന്നഡയിലും പത്താം തരത്തിലെ ഐ.ടി. ഒഴികെയുള്ള വിഷയങ്ങളിൽ ചോദ്യശേഖരം തയ്യാറാക്കി നൽകാനാണ് ഉദ്ദേശിക്കുന്നത്. ഈ അധ്യയനവർഷം പത്താംതരം പരീക്ഷ എഴുതുന്ന മുഴുവൻ കുട്ടികൾക്കും ഈ പദ്ധതിയുടെ പ്രയോജനം ലഭിക്കുമെന്ന് പ്രതീക്ഷിക്കുന്നു. വ്യത്യസ്ത പഠനശേഷിയും പഠനമികവുമുള്ള എല്ലാ വിഭാഗം കുട്ടികൾക്കും ഈ സാമഗ്രി പ്രയോജനപ്പെടട്ടെ എന്ന് ആശംസിക്കുന്നു. ഒരു കൂട്ടം അധ്യാപകരുടെ ആത്മാർത്ഥമായ പ്രവർത്തനമാണ് ഈ സാമഗ്രി വികസിപ്പിക്കാൻ സഹായകമായത്. അധ്യാപകരുടെ ആത്മാർത്ഥമായ പിന്തുണയും പ്രോത്സാഹനവും അനിവാര്യമായ ഈ ഉദ്യമത്തിൽ എല്ലാവരുടെയും സഹായസഹകരണങ്ങൾ തുടർന്നും പ്രതീക്ഷിച്ചുകൊണ്ട് ഏവർക്കും വിജയാശംസകൾ നേരുന്നു. ആശംസകളോടെ,

ഡോ. രഘുരാമ ഭട്ട് കെ.
പ്രിൻസിപ്പാൾ
ഡയറ്റ് കാസറഗോഡ്



ആമുഖം

കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ജില്ലയിലെ പൊതുവിദ്യാഭ്യാസം ശക്തിപ്പെടുത്തുന്നതിന് വ്യത്യസ്തങ്ങളായ നിരവധി പ്രവർത്തനങ്ങൾ നടന്നുവരികയാണ്. അതേസമയം ദേശീയ-സംസ്ഥാന പഠനങ്ങൾ നമ്മുടെ ജില്ലയിലെ കുട്ടികളുടെ പ്രകടനം ഇനിയും മെച്ചപ്പെടേണ്ടതുണ്ട് എന്ന സൂചനയാണ് നൽകുന്നത്. ഈ പശ്ചാത്തലത്തിലാണ് ജില്ലയിലെ പൊതുവിദ്യാലയങ്ങളിൽ നിന്ന് **2025-26** അധ്യയന വർഷം എസ്.എസ്.എൽ.സി. പരീക്ഷകൾ അഭിമുഖീകരിക്കുന്ന കുട്ടികൾക്ക് പഠനപിന്തുണ നൽകുന്നതിന് ജില്ലാ പഞ്ചായത്തിന്റെയും പൊതുവിദ്യാഭ്യാസ വകുപ്പിന്റെയും സംയുക്താഭിമുഖ്യത്തിൽ വ്യത്യസ്ത വിഷയങ്ങളിൽ പഠനസാമഗ്രികൾ തയ്യാറാക്കേണ്ടതിന്റെ ആവശ്യകത ജില്ലാതല ഉന്നതാധികാര യോഗങ്ങളിൽ ചർച്ചചെയ്യപ്പെട്ടത്. ഇതിന്റെ അടിസ്ഥാനത്തിൽ ഡയറ്റ് കാസറഗോഡിന്റെ അക്കാദമിക നേതൃത്വത്തിൽ ജില്ലയിലെ മികച്ച അധ്യാപകരെ ഉൾപ്പെടുത്തിക്കൊണ്ട് പത്താംതരത്തിൽ ഐ.ടി. ഒഴിച്ചുള്ള എല്ലാ വിഷയങ്ങളിൽ വിലയിരുത്തൽ സമീപനത്തിലനുസരിച്ച് ചോദ്യശേഖരം **(Question Pool)** തയ്യാറാക്കിയിരിക്കുകയാണ്. ജില്ലയിൽ നിന്നും പൊതുപരീക്ഷയെ അഭിമുഖീകരിക്കുന്ന മുഴുവൻ എസ്.എസ്.എൽ.സി, വിദ്യാർത്ഥികൾക്കും ആത്മവിശ്വാസം വളർത്തുന്നതിനും ഉന്നത വിജയം നേടുന്നതിനും ഈ ഉദ്യമം സഹായകമാകട്ടെയെന്ന് ആത്മാർത്ഥമായി ആഗ്രഹിക്കുന്നു. ഈ പദ്ധതിയെ നെഞ്ചേറ്റിയ പ്രിയപ്പെട്ട അധ്യാപക സുഹൃത്തുക്കൾക്ക് ഈ പുസ്തകത്തെ ഫലപ്രദമായി ഉപയോഗിക്കാൻ കഴിയട്ടെ.

എല്ലാവർക്കും വിജയാശംസകൾ.

ശ്രീ. മധുസൂദനൻ ടി. വി.
ജില്ലാ വിദ്യാഭ്യാസ ഉപഡയറക്ടർ
കാസറഗോഡ്

CHEMISTRY

English Medium

Chapter 1

Nomenclature of Organic Compounds & Isomerism

I Objective type, MCQ (1 mark)

1. If a H atom in a saturated hidrocarbon is replaced by -OH group, compound thus obtained is generally known as

   a) Ketone b) Alcohol

   c) Carboxylic Acid d) Ether

2. Match the Following

Functional group	Compounds
a) —COOH	(i) Aldehyde
b) —OH	(ii) Carboxylic Acid
c) —CHO	(iii) Ketone
d) $\diagup\!\!\diagdown C = O$	(iv) Alcohol

Choose the correct answer from the following

	(a)	**(b)**	**(c)**	**(d)**
(A)	(ii)	(i)	(iv)	(iii)
(B)	(iv)	(iii)	(ii)	(i)
(C)	(ii)	(iv)	(i)	(iii)
(D)	(iii)	(iv)	(ii)	(i)

3. Which compound is obtained, when the hydrogen (H) atom in the aromatic compound C_6H_6 is replaced by $-COOH$ group?

a) Phenol
b) Benzoic Acid
c) Hexanoid acid
d) Hexanol

4. Statement 1 : The poistion of functional group in the compound,

$CH_3 - CH_2 - CH_2 - CO - CH_3$ is 2.

Statement 2 : This Compound is an example of ethers.

Which of the following is correct regarding this statement.

a) Statement 1 is true, but statement 2 is wrong.
b) Both Statements are true.
c) The Statement 1 is wrong but statement 2 is correct.
d) Both Statements are wrong.

5. Compounds containing alkoxy group are generally called .....

a) Ketone
b) Aldehyde
c) Carboxylic Acid
d) Ether

6. Write the suffix of the compound given below.

$CH_3 - CH = CH - CH_3$

a) ene
b) yne
c) ane
d) ol

7. What are the position number of branches in the given hydrocarbon?

$$CH_3 - CH_2 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_2 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_3$$

a) (3, 5) b) (2, 5)

c) (3, 4) d) (2, 4)

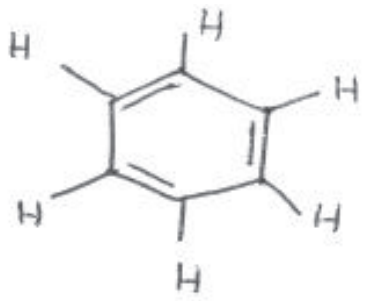
8. While giving the IUPAC name of a compound containing -CHO functional group, the word ..... is added at the end of the name.

a) ol b) al

c) one d) ane

9. The following is an organic compound.

This compound belongs to,



a) Aliphatic b) Alicyclic

c) Aromatic d) None of these

10. An Organic compound is given below.

$CH_3 - O - CH_2 - CH_2 - CH_2 - CH_3$

Which of the following is considered as alkoxy group while naming this compound.

a) Methoxy b) Ethoxy

c) Propoxy d) Butoxy

11. *Statement 1* : $CH_3 - CH_2 - O - CH_2 - CH_2 - CH_3$ is an example of ether.

Statement 2 : Its functional isomers are ketones.

Which of the following is correct?

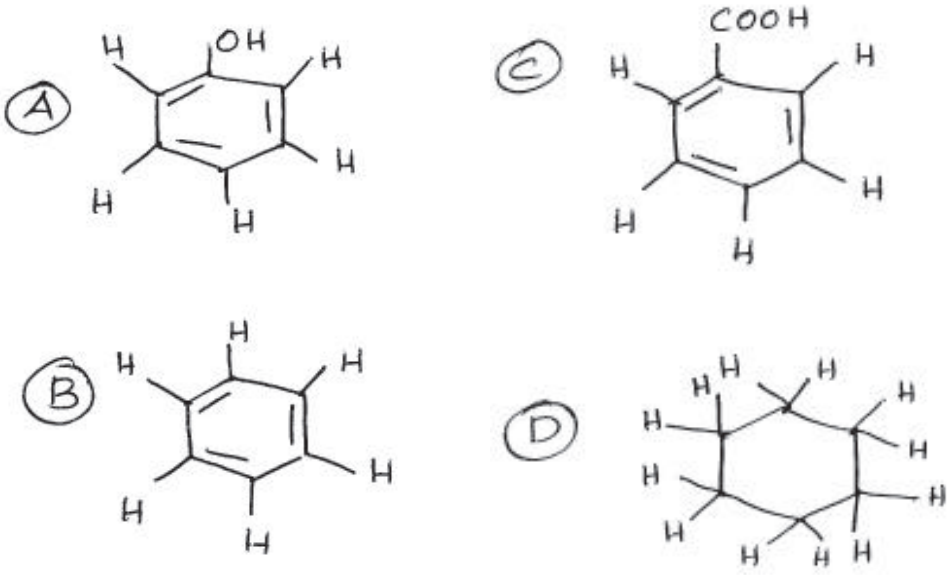
a) Both statements are correct

b) Both statements are wrong

c) The statement (1) is correct, but statement (2) is wrong

d) The statement (1) is wrong, but statement (2) is correct

12. Which of the following is not an aromatic Compound.



13. The Structural formulae of some compounds are given below.

Which of these are chain isomers?

i) $CH_3 - CH_2 - CH_2 - CH_3$

ii) $CH_3 - CH_2 - CH_2 - CHO$

iii) $CH_3 - CH_2 - CH_2 - OH$

iv) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_3$

a) i) and ii)

b) i) and iii)

c) ii) and iii)

d) i) and iv)

14. a) $CH_3 - CH_2 - CH = CH_2$

b) $CH_3 - CH = CH - CH_3$

Which of the following statements are correct?

i) Their general formula is C_nH_{2n+2}

ii) They are position isomers

iii) They belong to alkene family

iv) They are chain isomers

a) i) and ii) b) ii) and iii)

c) i) and iv) d) ii) and iv)

15. Which functional group does Methoxy ethane contain?

a) Aldehide group b) Hydroxyl group

c) Alkoxy group d) Keto group

16. Which of the following compounds have the same molecular formula?

i) $CH_3 - CH_2 - CHO$ ii) $CH_3 - CH_2 - CH_2 - OH$

iii) $CH_3 - O - CH_3$ iv) $CH_3 - CO - CH_3$

a) i) and ii) b) i) and iii)

c) ii) and iii) d) i) and iv)

17. a) $CH_3 - CH_2 - CO - CH_2 - CH_3$

b) $CH_3 - CO - CH_2 - CH_2 - CH_3$

c) $CH_3 - CH_2 - CH_2 - CO - CH_3$

d) $CH_3 - CH_2 - CO - CH_2 - CH_2 - CH_3$

Which of these compounds has the same alkyl groups on both sides of the functional group?

18. Which of the following are metamers?

i) $CH_3 - CH_2 - CH_2 - CH_2 - OH$ ii) $CH_3 - O - CH_2 - CH_2 - CH_3$

iii) $CH_3 - CH_2 - O - CH_2 - CH_3$ iv) $CH_3 - CH_2 - \underset{\displaystyle OH}{\underset{|}{CH}} - CH_3$

a) i) and ii) b) ii) and iii)

c) i) and iv) d) ii) and iv)

19. Which are the isomer pairs in the following compounds?

i) $CH_3 - CH_2 - CH_2 - OH$ ii) $CH_3 - CO - CH_2 - CH_3$

iii) $CH_3 - CH_2 - CH_2 - CHO$ iv) $CH_3 - O - CH_2 - CH_3$

a) i) and iii)
ii) and iv)

b) i) and iv)
ii) and iii)

c) i) and ii)
iii) and iv)

d) None of these

20. The IUPAC names of 4 compounds are given below. Which of these is not included in chain isomers with the molecular formula C_5H_{12}?

a) 2, 2 - dimethyl propane

b) Pentane

c) 2 - methyl pentane

d) 2 - methyl butane

21. Which among the following compounds does not have the molecular formula C_3H_8O?

a) Methoxy ethane b) Propan - 2 - ol

c) Propanal d) Propan - 1 - ol

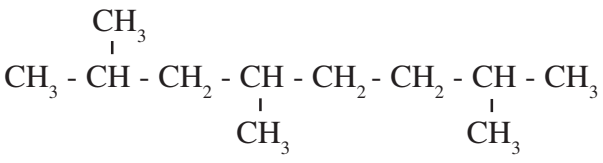
II. Very Short Answer type Questions (2 mark)

1. A saturated aliphatic hydrocarbon chain has a total of six carbon atoms and a methyl group attached to the second carbon atom in the main chain.

a) Draw the structure of this compound

b) Write the IUPAC name of this compound

2.



a) Find the position numbers of the branches in the IUPAC name of the compound given above.

b) Write the IUPAC name of this compound.

3. $CH_3 - CH_2 - C \equiv CH$

a) What is the correct position of the carbon containing the triple bond in the given organic compound?

b) What is the suffix of the IUPAC name of this compound?

4. Propane $\xrightarrow[\text{by Cl atom}]{\text{H atom replaced}}$ X

a) Write the structural formula of the compound X formed here.

b) Write the IUPAC name of this compound.

5. $CH_3 - CH_2 - CH_2 - CO - CH_3$

a) What is the correct position number of the functional group in the above compound?

b) Which category does it belong to?
(Ester, Ether, Ketone, Aldehyde)

6. The IUPAC name of 4 compounds are given below.

A) Methoxy propane B) Methoxy pentane

C) Ethoxy ethane D) Ethoxy propane

a) Which of these are metamers?

b) Write down their structural formulae?

7. i) $CH_3 - CH_2 - CH_2 - CH_2 - Cl$

ii) $CH_3 - \underset{\underset{Cl}{|}}{CH} - CH_2 - CH_3$

Two statements are given regarding the above compounds. Which of the following is the correct statement? Justify your answer.

Statement 1 : Both are functional isomers.

Statement 2 : Both are position isomers.

8. a) Which functional group does the given compound contain?

$CH_3 - COO - CH_2 - CH_2 - CH_3$

b) Write the structural formula of Carboxylic acid which has the same molecular formula as the above compound?

9. Write down the structural formula and IUPAC names of the metamer pairs with the molecular formula $C_5H_{12}O$.

10. An aliphatic organic compound has 3 carbon atoms in one molecule and is a saturated alcohol.

a) Write the structural formula of this compound?

b) Write the IUPAC name of the functional isomer of this compound.

11. Correct the incorrect statements in the following.

a) Isomers are molecules with the same structural formula but different molecular formula.

b) Propan - 1 - ol is the isomer of the compound $CH_3 - CH_2 - CHO$.

c) Popan - 1 - ol is the position isomer of the compound $CH_3 - \underset{\underset{OH}{|}}{CH} - CH_3$

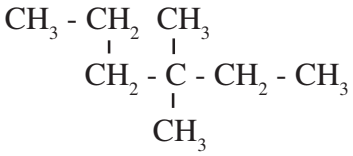
III Short Answer type Questions (3 mark)

1. The chemical formula of an organic compound is C_6H_{14}. It has a methyl group attached to the second Carbon atom and the third Carbon atom.

a) Draw the structural formula of this compound.

b) Write the IUPAC name of this compound.

c) Draw the structural formula of an isomer with 5 Carbon atoms in its mainchain.

2. 

a) How many Carbon atoms are in this longest carbon chain?

b) What is the position of the Carbon atom to which the methyl groups are attached?

c) Write the IUPAC name of the compound.

3. The molecular formula of a hydrocarbon is C_4H_8.

a) What type of hydrocarbon is this?

(Alkene, Alkyne, Alkane)

b) Draw the structure of an alicyclic compound with the same formula.

c) Write the IUPAC name of this compound.

4. A chain with five Carbon atoms is given below.

$C - C \equiv C - C - C$

a) Complete the structure by adding hydrocarbon atoms to each carbon.

b) Write the IUPAC name of this compound.

c) Write the structural formula of its position isomer.

5. The molecular formula of a compound is C_5H_{12}.

Write the structures and IUPAC names of the possible chain isomers of this compound.

6. i) $CH_3 - CH_2 - CH_2 - OH$ ii) $CH_3 - O - CH_2 - CH_3$

a) Write the functional group of each one

b) Write the IUPAC name of these.

c) What type of isomerism do these exhibit?

7. a) Write the IUPAC name of the compound, which is commonly known as acetone.

b) Which family of organic compounds contains this functional group?

c) Draw the structural formula of a compound that has the same molecular formula as this compound but a different functional group.

8. The structural formula of a compound is given below.

$$CH_3 - CH_2 - \underset{\displaystyle Br}{\underset{|}{CH}} - CH_2 - CH_3$$

a) Write the IUPAC name of this compound.

b) Write the structural formula and IUPAC name of all possible position isomers of this compound without changing the carbon chain.

IV. Short Answer type Questions(4 mark)

1. The Molecular formula of a hydrocarbon A is C_4H_{10}.

a) Which kind of hydrocarbons are these?

b) Draw the structural formula of the compound (B) obtained if one hydrogen atom is replaced by - OH.

c) Write the IUPAC name of 'B'

d) Draw the structural formula of the compound C that belongs to the same homologous series of B with two carbon atoms.

2. The functional group of a compound having the molecular formula $C_3H_6O_2$ is –COOH.

   a) This compound is belongs to which category?

      (Alcohol, Ester, Carboxylic acid)

   b) Write the structural formula of this compound?

   c) Write the IUPAC name of the compound.

   d) Give the IUPAC name of the compound obtained when this compound react with ethanol in presence of Conc. H_2SO_4.

3. The structural formula of the compound A is given below.

   $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - CHO$

   a) Write the IUPAC name of the compound.

   b) B is a functional isomer of compound A.

      Write the structural formula and IUPAC name of B.

   c) If C is a metamer of the compound B. Write the structure of C.

4. Find the isomer pairs from the compounds given below and write what type of isomers they are.

   a) $CH_3 - CO - CH_2 - CH_3$

   b) $CH_3 - CH_2 - CH_2 - OH$

   c) $CH_3 - CH_2 - CH_2 - CH_2 - Cl$

   d) $CH_3 - CH_2 - CH_2 - CHO$

   e) $CH_3 - CH_2 - CO - CH_2 - CH_3$

   f) $CH_3 - O - CH_2 - CH_3$

   g) $CH_3 - CO - CH_2 - CH_2 - CH_3$

   h) $CH_3 - CH_2 - \underset{\displaystyle Cl}{\underset{|}{CH}} - CH_3$

5. IUPAC name of a compound is 2 - methyl hexane.

a) Write the structural formula and molecular formula of the above compound.

b) How many Carbon atoms are present in the mainchain.

c) Write the structural formula of the chain isomer of this molecular with four Carbon atoms in the mainchain.

d) Write the IUPAC name of the above isomer.

Chapter 1

Nomenclature of Organic Compounds & Isomerism

ANSWER KEY

Objective type - 1 mark

1. b
2. C
3. b
4. a
5. d
6. a
7. d
8. b
9. c
10. a
11. c
12. d
13. d
14. b
15. c
16. d
17. a
18. b
19. b
20. c
21. c

Very Short Answer - 2 mark

1. a) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_2 - CH_2 - CH_3$

   b) 2 - methyl pentane

2. a) 2, 4, 7

   b) 2, 4, 7 - trimethyl octane

3. a) 1

   b) yne

4.a) $CH_3 - CH_2 - CH_2\,Cl$

   or

   $CH_3 - \underset{\displaystyle Cl}{\underset{|}{CH}} - CH_3$

   Any one of the above.

   b) 1 - chloropropane Or
   2 - chloropropane

5. a) 2

   b) ketones

6. a) A and C

   b) $A \Rightarrow CH_3 - O - CH_2 - CH_2 - CH_3$

   $C \Rightarrow CH_3 - CH_2 - O - CH_2 - CH_3$

7. Both are position isomers.

   Reason : Functional group is same. Position of the functional group (Cl) is different.

8. a) $-COO-$

b) $CH_3 - CH_2 - CH_2 - CH_2 - COOH$

9. $CH_3 - O - CH_2 - CH_2 - CH_2 - CH_3$: Methoxy butane

$CH_3 - CH_2 - O - CH_2 - CH_2 - CH_3$: Ethoxy Propane

10. a) $CH_3 - CH_2 - CH_2 - OH$

or

$$CH_3 - \underset{\displaystyle OH}{\underset{|}{CH}} - CH_3$$

b) methoxy ethane

11. a) Isomers are molecules with the same molecular formula but different structural formula.

b) Propan - 1 - ol is the isomer of the compound $CH_3 - O - CH_3$

or

Propanone is the isomer of the compound $CH_3 - CH_2 - CHO$

Short Answer (3 mark)

1. a)

$$CH_3 - \underset{\displaystyle CH_3}{\underset{|}{CH_3}} - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_3$$

b) 2, 3 - dimethyl butane

c)

$$CH_3 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_2 - CH_2 - CH_3$$

Or

$$CH_3 - CH_2 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_2 - CH_3$$

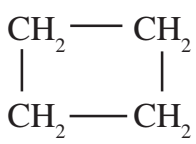
2. a) 6

b) 3

c) 3, 3 - dimethyl hexane

3. a) Alkene

b)



c) cyclobutane

4. a) $CH_3 - C \equiv C - CH_2 - CH_3$

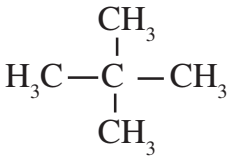
b) pent - 2 - yne

c) $HC \equiv C - CH_2 - CH_2 - CH_3$

5. $CH_3 - CH_2 - CH_2 - CH_2 - CH_3 \Rightarrow$ pentane



$\Rightarrow$ 2 - methyl butane



$\Rightarrow$ 2, 2 - dimethyl propane

6. a) i) —OH (hydroxyl group)

ii) —O— R (alkoxy group)

b) i) propan - 1 - ol

ii) methoxy ethane

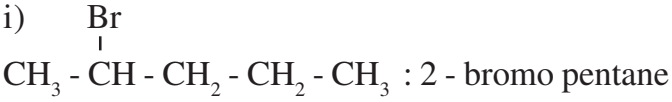
c) functional isomerism

7. a) Propanone

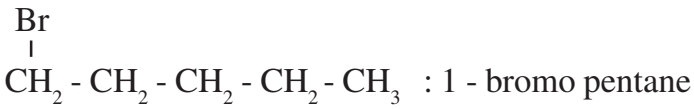
b) ketones

c) $CH_3 - CH_2 - CHO$

8. a) 3 - bromo pentane

b) i) 

: 2 - bromo pentane

ii) 

: 1 - bromo pentane

Short Answer - 4 mark

1. a) Alkane

b) $CH_3 - CH_2 - CH_2 - CH_2 - OH$ Or



c) butan - 1 - ol Or

butan - 2 - ol

d) $CH_3 - CH_2 - OH$

2. a) Carboxylic Acid

b) $CH_3 - CH_2 - COOH$

c) propanoic acid

d) Ethyl propanoate

Note : $CH_3 - CH_2 - COOH + CH_3 - CH_2 - OH \xrightarrow{Conc.H_2SO_4} CH_3 - CH_2 - COO - CH_2 - CH_3$

3. a) Hexanal

b) $CH_3 - CO - CH_2 - CH_2 - CH_2 - CH_3 \Rightarrow$ Hexan - 2 - one

Or

$CH_3 - CH_2 - CO - CH_2 - CH_2 - CH_3 \Rightarrow$ Hexan - 3 - one

c) $CH_3 - CH_2 - CO - CH_2 - CH_2 - CH_3$ Or

$CH_3 - CO - CH_2 - CH_2 - CH_2 - CH_3$

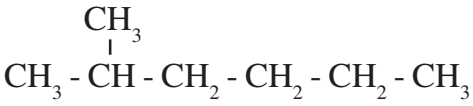
4. a & d : Functional isomers

b & f : Functional isomers

c & h : Position isomers

e & g : Metamers

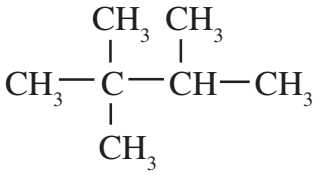
5. a)



(C_7H_{16})

b) 6

c)



d) 2, 2, 3 - trimethyl butane

Chapter 2

CHEMICAL REACTIONS OF ORGANIC COMPOUND

I Objective type, MCQ (1 mark)

1. Methane reacts with Chlorine in the presence of sunlight. This reaction is an example for .....

   a) Addition reaction b) Substitution reaction

   c) Polymerisation d) Thermal cracking

2. Which of the following compounds does not undergo addition reaction?

   a) C_2H_4 b) C_2H_2

   c) C_2H_6 d) C_3H_6

3. Compound 'X' reacts with a hydrogen molecule to form ethene. Which of the following compounds is likely to be 'X'?

   a) C_2H_6 b) C_2H_2

   c) C_3H_6 d) C_4H_8

4. Which of the following polymers is used to make the inner coating of non-stick cookware?

   a) Polytetrafluoroethene b) Polyvinyl Chloride

   c) Polythene d) Polyacrylonitrile

5. Which of the following is an example for condensation polymer?

a) PVC b) Teflon

c) Polythene d) Polyester

6. When a hydrocarbon 'A' containing 3 Carbon atoms and 8 hydrogen atoms undergoes thermal decomposition, methane and compound 'B' are formed. Identify 'A' and 'B' and select the correct pair.

a) A $\Rightarrow$ $CH_3 - CH_2 - CH_3$, B $\Rightarrow$ $CH \equiv CH$

b) A $\Rightarrow$ $CH_3 - CH_2 - CH_3$, B $\Rightarrow$ $CH_2 = CH_2$

c) A $\Rightarrow$ $CH_3 - CH = CH_2$, B $\Rightarrow$ $CH_2 = CH_2$

d) A $\Rightarrow$ $CH_3 - C \equiv CH$, B $\Rightarrow$ $CH_3 - CH_3$

7. Choose the correct statements from given below.

i) Adipic acid and Pentamethylene diamine are the monomers of nylon 66.

ii) Adipic acid and hexametlyene diamine are the monomers of nylon 66.

iii) Nylon 66 is a condensation polymer.

iv) Nylon 66 is an addition polymer.

a) i) & ii) b) ii) & iii

c) iii) & iv) d) i) & iii)

8. Statement 1 : Hydrocarbon on combustion forms CO_2 & H_2O.

Statement 2 : Combustion of hydrocarbon is an endothermic process.

a) Only statement 1 is true

b) Only statement 2 is true

c) Both the statements are true

d) Both the statements are false

9. Match the following.

A	B
(w) Wash	(p) 95.6% ethanol
(x) Rectified spirit	(q) 100% ethanol
(y) Absolute alcohol	(r) used as fuel
(z) Power alcohol	(s) 8-10% ethanol

Choose the correct answer.

	w	x	y	z
a)	s	q	p	r
b)	s	p	r	q
c)	s	p	q	r
d)	p	q	r	s

10. Which organic compound is prepared by the fermentation of diluted molasses?

a) Ethanoic acid b) Ethanol

c) Vinegar d) Ethyl ethanoate

11. Which chemical reaction is used for controlling plastic pollution?

a) Substitution reaction b) Addition reaction

c) Combustion d) Thermal cracking

12. The polymer orlon is used to make synthetic fibres. The monomer present in orlon is?

a) Vinyl cyanide b) Vinyl chloride

c) Ethene d) Tetra fluoro ethene

13. What percentage of ethanoic acid is vinegar?

a) 5-8% b) 8-10%

c) 10-12% d) 5-10%

14. What are the reactants required for the preparation of methyl propanoate?

a) Methanol, Propanoic acid

b) Propanol, Methanoic acid

c) Methanol, Ethanoic acid

d) Ethanol, Ethanoic acid

15. Match the following.

A	B
(X) Antipyretic	(p) Pencillin
(Y) Antibiotic	(q) Detol
(Z) Antiseptic	(r) Aspirin
	(s) Paracetamol

	X	Y	Z
a)	p	q	r
b)	q	r	s
c)	s	p	q
d)	r	p	q

II Very Short Answer type Questions (2 mark)

16. a) Write the chemical equation for the formation of the ester, when methanol reacts with ethanoic acid.

b) Write anyone use of esters?

17. a) What is esterification?

    b) Write the name of the ester present in Jasmine?

18. a) Explain the industrial preparation of methanol.

    b) Write anyone use of methanol?

19. From the chemical reactions given below, write the steps in correct order for the industrial production of ethanol.

    A) Invertase converts sucrose to glucose & fructose

    B) Yeast is added to diluted molasses.

    C) Zymase converts glucose & fructose to wash

    D) Wash on fractional distillation forms ethanol

20. Carbon Monoxide reacts with methanol in the presence of catalyst produces 'A'?

    a) Identify the compound 'A'?

    b) Write the chemical equation of this reaction?

21. 95.6% ethanol is called rectified spirit.

    a) Why rectified spirit is denatured?

    b) Methanol added ethanol is called .....

22. 'X' on thermal decomposition forms ethane & ethene.

    a) Write the structure of 'X'?

    b) Write anyone factor which influence thermal cracking.

23. a) What are analgesics?

b) Which medicine is used as a pain reliever and to prevent heart attack?

24. a) $\underline{\text{(A)}} + H_2 \rightarrow CH_2 = CH_2$

b) $CH_2 = CH_2 + HBr \rightarrow \underline{\text{(B)}}$

Identify A, B and write their structure.

25. a) Define Polymerisation.

b) Based on the method of polymerisation there are two type of polymers. What are they?

III Short Answer type Questions (3 mark)

26. $CH \equiv CH + H_2 \rightarrow \underline{\text{A}}$

a) Identify 'A'?

b) Write the products obtained when 'A' undergoes the following reactions.

i) $A + HCl \rightarrow \underline{\text{B}}$

ii) $nA \rightarrow \underline{\text{C}}$

27. a) Which of the following is not an example for addition polymer?

(PVC, Teflon, Nylon 66)

b) Write the structural formula of the monomers of nylon.

28. When ethyl alcohol reacts with a carboxylic acid containing two carbon atoms in the presence of concentrated H_2SO_4, a fragrant compound obtained.

a) Common name of the product formed is?

b) Write the Chemical equation for this reaction?

c) Write the IUPAC name of the product.

IV Short Answer type Question (4 mark)

29. a) Identify 'A' & 'B' in the following reactions.

i) $\Rightarrow 2CH_3 - CH_2 - CH_2 - CH_3 + 13O_2 \rightarrow \underline{\quad 8(A) \quad} + 10H_2O + \text{Heat}$

ii) $\Rightarrow C_4H_{10} \xrightarrow{\text{Heat}} \underline{\quad (B) \quad} + CH_4$

b) Write the difference between combustion and thermal cracking?

c) CH_4 is used as a fuel. Give reason.

30. $C_{12}H_{22}O_{11} + H_2O \xrightarrow{\text{Invertase}} C_6H_{12}O_6 + \underline{\quad (A) \quad}$

$C_6H_{12}O_6 \xrightarrow{\text{Zymase}} \underline{\quad (B) \quad} + CO_2$

a) Identify A & B?

b) How rectified spirit is obtained from wash?

c) What is power alcohol?

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Chapter 2

CHEMICAL REACTIONS OF ORGANIC COMPOUND

ANSWER KEY

Objective type (1 mark)

1. b)
2. c)
3. b)
4. a)
5. d)
6. b)
7. b)
8. a)
9. c)
10. b)
11. d)
12. a)
13. a)
14. a)
15. c)

Very Short Answer (2 mark)

16. a) $CH_3 - OH + HOOC - CH_3 \xrightarrow{\text{Con. } H_2SO_4} CH_3 - OOC - CH_3 + H_2O$

    b) To make artificial perfumes or
    To make artificial drinks

17. a) The reaction between an alcohol and a carboxylic acid to form esters is called esterification.

    b) Benzyl ethanoate

18. a) Methanol is industrially prepared by the reaction between CO & H_2 in the presence of a catalyst.

b) To make varnish & paint

(To make formic acid, To make formaldehyde)

19. B, A, C, D

20. a) CH_3 - COOH - Ethanoic acid

b) $CO + CH_3\text{-}OH \xrightarrow[573K]{\text{Catalyst}} CH_3 - COOH$

21. a) To avoid alcohol abuse

b) Methylated spirit

22. a) $CH_3 - CH_2 - CH_2 - CH_3$

b) Temperature/Pressure/Nature of hydrocarbon

23. a) Medicines which are used as pain relievers

b) Aspirin

24. $A \Rightarrow CH \equiv CH$

$B \Rightarrow CH_3 - CH_2 - Br$

25. a) The process by which many simple moleculer (monomers) combine to form complex moleculer is called polymerisation.

b) Addition Polymers, Condensation poymers.

Very Short Answer (3 mark)

26. a) $A \Rightarrow CH_2 = CH_2$

b) i) $B \Rightarrow CH_3 - CH_2 - Cl$

ii) $C \Rightarrow \left[CH_2 - CH_2\right]_n$

27. a) Nyon 66

b) $NH_2\{CH_2\}_6 NH_2$, $HOOC\ \{CH_2\}_4 COOH$

28. a) Ester

b) $CH_3 - COOH + HO - CH_2 - CH_3 \xrightarrow{Con.\ H_2SO_4} CH_3 - COO - CH_2 - CH_3 + H_2O$

c) Ethylethanoate

Short Answer (4 mark)

29. a) $A \Rightarrow CO_2 \quad B \Rightarrow CH_3 - CH = CH_2$

b) Combustion $\Rightarrow$ Chemical reaction taking place in the presence of O_2

Thermal Cracking $\Rightarrow$ Chemical reaction taking place in the absence of O_2

c) Combustion of methane produces large amount of heat and energy

30. a) $A \Rightarrow C_6H_{12}O_6$

$B \Rightarrow C_2H_5OH$

b) By Fractional distillation

c) Mixture of 20% absolute alcohole & 80% petrol.

Chapter 3

PERIODIC TABLE AND ELECTRON CONFIGURATION

I Objective type, MCQ (1 mark)

1. Which orbital is filled after 3p according to the $(n+\ell)$ rule?

   a) 3d b) 4s

   c) 4p d) 3f

2. Match the folowing

Quantum Number	Description
a) Principal Quantum Number (n)	1) Orientation of Orbital
b) Azimuthal Quantum Number (ℓ)	2) Shell Number
c) Magnetic Quantum Number (m)	3) Shape of orbital

   a) a - 1, b - 2, c - 3 b) a - 3, b - 1, c - 2

   c) a - 2, b - 2, c - 1 d) a - 2, b - 3, c - 1

3. Statement 1 : The electron configuration of Calcium is $[Ar]4s^2$.

   Statement 2 : Calcium belongs to period 4 and group 2.

   Which one of the following is correct?

   a) Both statements are correct

   b) Both statements are incorrect

c) Only Statement 1 is correct.

d) Only statement 2 is correct

4. Which of the following pairs of subshells have the same $(n+\ell)$ value? Choose the correct answer from the given option?

a) 2p, 3s b) 3d, 4s

c) 3s, 4p d) 4p, 5s

A) a) and c) B) a) and d)

C) c) and d) D) b) and c)

5. The total number of orbitals present in the 'N' shell is,

a) 16 b) 32

c) 8 d) 18

6. Which of the following element has highest ionisation enthalpy?

a) Lithium b) Oxygen

c) Fluorine d) Neon

7. Which block does the element with subshell electron configuration [Ne]$3s^2$ $3p^5$ belong to?

a) s - bock b) p - Block

c) d - block d) f - block

8. Which of the following has the lowest ionisation enthalpy?

a) $1s^2$ $2s^2$ $2p^6$ b) $1s^2$ $2s^2$ $2p^5$

c) $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^1$ d) $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^1$

9. Which subshell has the highest energy among the following?

a) 3p b) 3d

c) 4s d) 2p

10. The electron configuration of Copper is [Ar] $3d^{10}$ $4s^1$ instead of [Ar] $3d^9$ $4s^2$. The reason is,

a) Full-filled 'd' subshell stability

b) Half filled 'd' subshell stability

c) Full fillled 's' subshell stability

d) Half filled 's' subshell stability

11. Which of the following period has the maximum number of elements?

a) 2nd period b) 3rd period

c) 5th period d) 6th period

12. Match the following

Scientist	Discovery
a) Louis de Broglie	1. Uncertainity principle
b) Heisenberg	2. Stationary energy levels
c) Neils Bohr	3. Wave nature of matter

Choose the correct answer

a) a - 2, b - 3, c - 1 b) a - 3, b - 1, c - 2

c) a - 1, b - 2, c - 3 d) a - 3, b - 2, c - 1

13. There is an element in 's' block which exist in solid state in winter. But if you hold it on your palm it will melt. Name the element?

a) Caesium b) Gallium

c) Mercury d) Bromine

14. Which among the following compounds are coloured?

a) $CaCO_3$ b) $KMnO_4$ c) $ZnSO_4$ d) $K_2Cr_2O_7$

a) a and b b) b and c

c) b and d d) c and d

15. Which of the following elements are used on computers, LCD Screens, Mobile phones, Batteries etc

a) Actinoids b) Rare earth elements

c) Transition Elements d) Main group elements

II Short Answer type Questions (2 mark)

1. The atomic numbers of elements A and B are 20 and 17 respectively, Then,

a) Write the subshell electron configuration of A and B?

b) Write the molecular formula of compound formed by combining A and B?

2. The subshell electron configuration of an element 'X' is $[Ar]3d^6\ 4s^2$.

a) Find the group number.

b) Find the period and block.

3. a) Define ionisation enthalpy?

b) How does it vary down a group?

4. Write the n, ℓ values of the electron in '3p' subshell.

5. a) Arrange the following subshell in the increasing order of energy?

3s, 3p, 4s, 3d, 2p

b) Which among the following has the highest $(n + \ell)$ value?

(i) 5s (ii) 4d

6. The Sub shell electronic configuration of two elements are,

A - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ B - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$

a) Write the formula of the compound formed by combining A and B?

b) Whether the compound is ionic or covalent?

7. One subshell is described by quantum number $n = 3, \quad \ell = 2$.

   a) Name the subshell?

   b) Write the maximum number of electrons that can be accommodated in the 4d sub shell?

8. a) Find the Oxidation state of Fe in the compounds given below.

   (i) $FeCl_2$ ii) $Fe\ Cl_3$

   b) Why does Fe exihibit variable oxidation state?

9. a) In which of the following compounds, Manganese exihibit +3 Oxidation State.

   (P) $MnCl_2$ (Q) MnO_2 (R) Mn_2O_3 (S) Mn_2O_7

   b) Write the subshell electron configuration of Manganese ion present in Mn_2O_7.

10. The last electron of element 'X' is in the p subshell. The element 'X' belongs to 3rd period and 17th group.

    a) Write the subshell electron configuration of 'X'.

    b) Write the subshell electron configuration of 'Y' which is in the same group above the element 'X' in the periodic table.

III Short Answer type Questions (3 marks)

1. The Principal quantum number of the outermost electron in the atom is n = 3.

   a) What is the total number of orbitals in this atom?

   b) How do you denote the subshell of this shell where $\ell = 2$?

   c) How many orientations are possible in this sub shell?

2. The outermost subshell of an atom with principal quantum number 2 contains five electrons.

   a) Write sub shell electron configuration of this atom.

b) What are the subshells of the shell when n = 2.

c) How many orientations are possible for a sub shell with $\ell = 0$.

3. Lanthanoids and actinoids belongs to f - block.

a) Which element is used for making strong magnet?

b) Write the use of Cerium and Lanthanum?

c) Write the name of elements which are used as fuel in nuclear reactors?

4. Rare earth elements consists of 17 elements including 15 elements of Lanthanoids.

a) Which are the other two elements included as rare earth?

b) One of the main ore of rare earth elements is commonly found in coastal regionsof Southern Kerala. Name it.

c) Mention the uses of Rare earth elements in the field of technology?

5. There are 4 blocks in the periodic table.

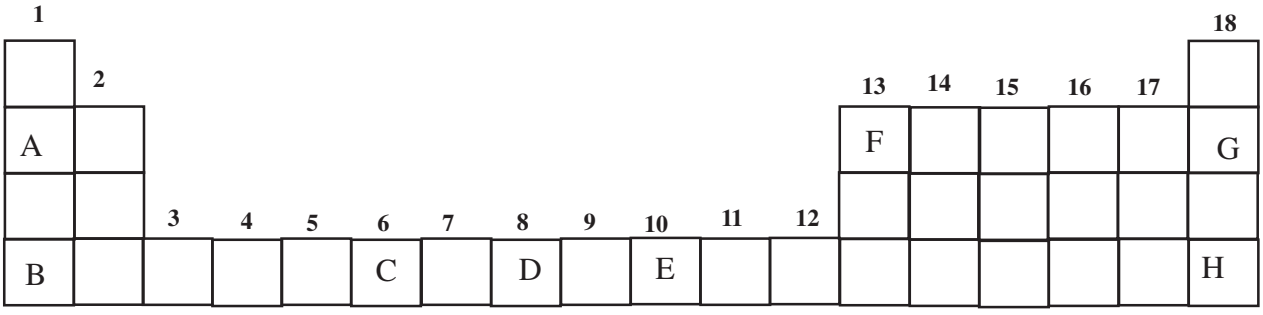
a) Name the block which includes groups 3 to 12.

b) Write any two characteristics of the elements in this block.

IV Short Answer type Questions (4 mark)

1. A portion of the periodic table is given. Answer the questions given below.

(Symbols are not real)



a) Which element has the highest ionisation enthalpy?

b) Sub shell electron configuration of C is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$. Then write the electronic configuration of E.

c) Which element in the 2nd period has the highest nuclear charge?

d) Which element has only one electron in the outermest 'p' Subshell?

2. The Sub shell electron configuration of a few elements are written on the basis of noble gas.

(Symbols are not real)

i) P - $[Ar]\ 4s^2$ ii) Q - $[Ne]\ 3s^2\ 3p^1$

iii) R - $[Ne]\ 3s^2\ 3p^5$ iv) S - $[Ar]\ 3d^6\ 4s^2$

a) Which element forms coloured compound?

b) Write the chemical formula of the compound formed , When element 'P' reacts with element 'R'.

c) Which element has the highest ionisation enthalpy?

d) Write the subshell electron configuration of the noble gas which is in the same period as element 'Q'.

3. Manganese belongs to 'd' block.

a) Atomic number of Manganese is 25. Write its subshell electron configuration based on noble gas.

b) Find the Oxidation number of Manganese in $MnCl_2$?

c) Write the sub shell, configuration of Mn^{4+}.

d) Why does 'd' block elements show variable Oxidation state?

4. a) What significant discovery did Louis de Broglie make regarding the nature of Matter?

b) What does Heisen berg's uncertainity Principle state about subatomic particles like electrons?

c) Write the major limitation of the Bohr model of the atom?

d) Where are the electrons most likely to be seen according to the quantum mechanical model?

5. The maximum number of electrons in a subshell is given by $2(2\ell+1)$.

a) Calculate the maximum number of electrons in 'd' subshell.

b) Find the total number of orbitals present in the 'f' subshell?

c) Verify the possibility of a subshell with quantum number values $n=2$, $\ell=2$?

Chapter 3

PERIODIC TABLE AND ELECTRON CONFIGURATION

ANSWER KEY

Objective type (1 mark)

1. b)
2. d)
3. a)
4. B)
5. a)
6. d)
7. b)
8. c)
9. b)
10. a)
11. d)
12. b)
13. a)
14. c)
15. b)

Short Answer (2 mark)

1. a) A - 20 - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$

   B - 17 - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

   b) AB_2

2. a) 8

   b) Period - 4, block - d

3. a) Ionisation enthalpy is the minimum amount of energy required to remove the most loosely bound electron from the outermost shell of an isolated gaseous atom of an element.

   b) Ionisation enthalpy decreases down a group.

4. 3p

n	ℓ
3	1

5. a) 2p, 3s, 3p, 4s, 3d

   b) 4d

6. a) BA

   b) Ionic

7. a) 3d

   b) 10

8. a) i) +2, ii) +3

   b) In transition elements there is only a slight energy difference between the outermost 's' subshell and the penultimate 'd' subshell. So electrons from the 'd' subshell also take part in chemical reaction.

9. a) (R) Mn_2O_3

   b) Mn^{+7} - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

10. a) $_{17}X$ - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

    b) Y - $1s^2\ 2s^2\ 2p^5$

Short Answer (3 mark)

1. a) $n^2=9$

   b) 3d

   c) $2\ell+1=5$

2. a) $1s^2\ 2s^2\ 2p^5$

   b) s, p

   c) $2\ell+1=1$ (ie. s)

3. a) Neodymium

   b) They are used in catalyst in petroleum industry.

   c) Uranium, Thorium, Plutonium

4. a) Scandium and Yittrium

   b) Monazite

   c) Computers, LCD Screens, Mobile phones, Batteries

5. a) 'd' - block

   b) i) Variable Oxidation state

   ii) Forms coloured compounds

   iii) The electrons are being gradually filled up in the penultimate shell.

   iv) All are metals.

Short Answer (4 mark)

1. a) G

   b) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^8\ 4s^2$

   c) G

   d) F

2. a) iv) S [Ar] $3d^6\ 4s^2$

   b) PR_2

   c) R - [Ne] $3s^2\ 3p^5$

   d) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

3. a) [Ar]$3d^5\ 4s^2$

   b) +2

   c) Mn^{4+} - $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3$

   d) In transition elements there is only a slight energy difference between the outer most 's' subshell and the penultimate 'd' subshell. Hence 'd' subshell also take part in chemical reactions.

4. a) The wave nature of matter.

   b) It is impossible to determine simultaneously exact position and exact velocity of a fast moving subatomic particle like electron.

   c) On the basis of wave particle dual nature and uncertainity principle an electron cannot be considered as a particle moving along an orbit.

   d) The regions around the nucleus where there is maximum probability of finding the electrons (orbital)

5. a) $2(2\ell+1)$

Value of ℓ for d = 2

2(2x2+1) = 10

b) $2\ell+1 = 2x3+1 = 7$

c) When n=2, the possible value of ℓ are 0 and 1 only.

Chapter 4

GAS LAWS AND MOLE CONCEPT

I Objective type Multiple Choice Questions (1 mark)

(Choose only the most correct answer from A, B, C, D and write it down)

1. Which of the following statements is correct regarding the arrangement of molecules in the gasseous state as compared to the solid and liquid states?

   a) The distance between molecules is greatest in gases.

   b) The freedom of movement of molecules is least in the gasseous state.

   c) Force of attraction between molecules is least in gasseous state.

   d) Energy of molecules are comparatively low in gasseous state.

   A) a) and b) B) c) and d)

   C) a) and c) D) b) and d)

2. Which of the following is not one of the fundamental principles of Kinetic Molecular theory regarding the behaviour of gases?

   a) The Volume of gas can be reduced by increasing the distance between the molecules.

   b) Gas molecules are constantly moving in all directions.

   c) Average Kinetic energy of gasseous molecules are inversely proportional to Temperature.

   d) Collisions between gas molecules are elastic.

A) a) and b)

B) a) and c)

C) b) and c)

D) b) and d)

3. If CO_2 gas filled in a bottle of 5L capacity is transfered to a glass jar of 10L capacity and the jar is closed, what will be the current volume of the gas?

A) 5 L

B) 10 L

C) less than 5 L

D) More than 10 L

4. From the given substances, identify the elements which are in the gasseous state at normal temperature?

A) Oxygen, Caesium, Chlorine

B) Nitrogen, Hydrogen, Oxygen

C) Carbondioxide, Chlorine, Hydrogen

D) Chlorine, Gallium, Nitrogen

5. Which of the following is the correct relationship between some units for expressing volume?

i) $1m^3 = 100L$ ii) $1\ L = 100\ cm^3$

A) Both i and ii

B) Only i

C) Only ii

D) Bothe i and ii are wrong

6. Match the following

Measurement	SI Unit
1) Volume	a) K
2) Pressure	b) m^3
3) Temperature	c) Pa
4) Amount of substance	d) mol

	1	2	3	4
A)	a	b	d	c
B)	a	c	b	d
C)	b	d	c	a
D)	b	c	a	d

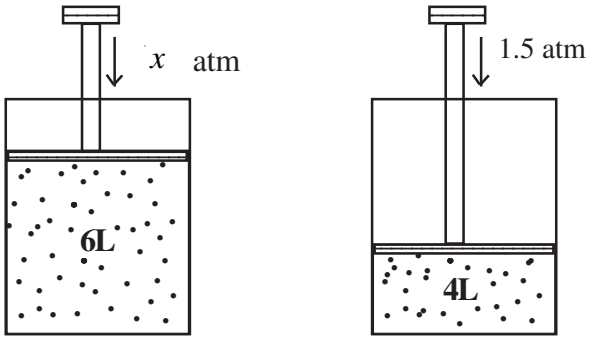
7. Melting point of Sodium is approximately 97.8^0C.

Express this in Kelvin Scale.

A) 100K B) 380.8K

C) 370.8K D) 380.7K

8.



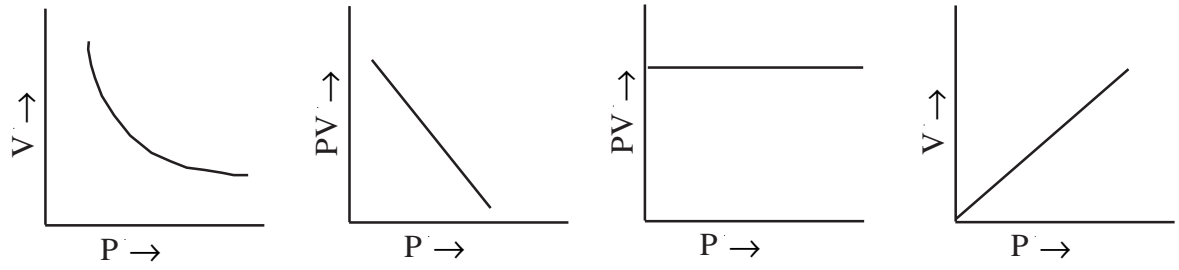
Situation 1 Situation 2

These two cases illustrated above involve applying pressure to a gas stored in a cylinder using a piston without changing its temperature. Calculate the pressure (x) experienced in the first situation?

A) 1.5 atm B) 2 atm

C) 1 atm D) 0.5 atm

9. Observe the graphs showing the relationship between volume and pressure of a given mass of gas without changing its temperature, and select only the correct ones.



(a) (b) (c) (d)

A) Only (a) B) (b) and (c)

C) (a) and (c) D) Only (d)

10. 1 atm = ... Pa

A) 1.01325×10^{6} B) 1.01325×10^{5}

C) 1.01325×10^{23} D) 1.01325×10^{10}

11. Relationship between volume and pressure of a fixed mass of gas at constant temperature is given below. Which is not correct?

A) $P\alpha\frac{1}{V}$

B) PV = a Constant

C) $P_1/V_1 = P_2/V_2$

D) $P_1V_1 = P_2V_2$

12. The volume and pressure of a given mass of gas at constant temperature are measured and listed in various cases.

Find 'x' and 'y'?

P (atm)	V(L)
2	6
1.5	x
y	4

A) $x = 8,\ y = 2$ B) $x = 3,\ y = 8$

C) $x = 8,\ y = 3$ D) $x = 5, y = 3$

13. Two statements related to the properties of gasseous molecules are given below.

Statement 1 : Gas exerts pressure

Statement 2 : Gasseous molecules collide with the walls of the container and there exert some force.

A) Statement 1 and 2 are false

B) Statement 1 is false and 2 is true

C) Both the statements are correct. Statement 2 is the correct explanation of statement 1.

D) Both the statements are correct. Statement 2 is not the correct explanation of statement 1.

14. Some observations in day to day life related to gas laws are given below.

1) The size of weather balloons increases as they rise above sea level.

2) During the summer, vehicle tyres are filled with air at low pressure.

3) The size of football increases when fill it with air.

4) Bottles filled with liquid ammonia are placed in cold water for a short time before opening.

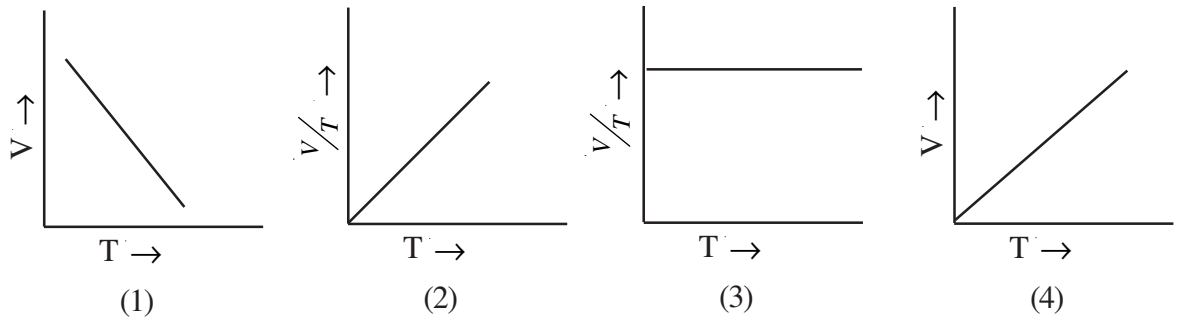
5) Blowing up a balloon

6) The bubbles rising to the top of the aquarium are getting bigger.

Classify which gas laws there are related to:

	Boyle's law	Charles law	Avagadro's law
A.	(6) and (3)	(1) and (4)	(2) and (5)
B.	(1) and (6)	(2) and (4)	(3) and (5)
C.	(1) and (2)	(3) and (4)	(5) and (6)
D.	(1) and (6)	(2) and (5)	(3) and (4)

15. Choose only the correct graph that shows how the Volume and Temperature in Kelvin Scale of fixed mass of gas are related when the pressure is constant.



(1) (2) (3) (4)

A) (1) and (3) B) (3) and (4)

C) (2) and (4) D) (2) and (3)

16. What happens to volume, when temperature of fixed mass of gas is increased at constant pressure:

A) Increase

B) Decrease

C) First decreases then increases

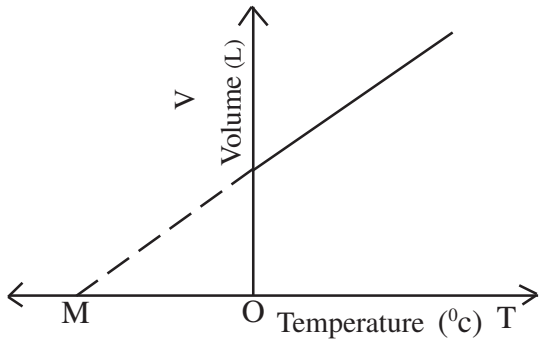
D) No change

17. The lowest imaginary temperature that a gas is to be able to reach

A) -273.15K B) 0^0C

C) 0 K D) 273.15^0C

18.



V - T graph related to volume and temperature of fixed mass of gas at constant pressure is given above. The temperature indicated by the point marked 'M' is?

A) Melting point of water B) Absolute Zero

C) Boiling point of water D) Normal atmospheric temperature

19. Charles law states the relationship between the volume and temperature of a gas.

1) $V \alpha T$

2) $V \alpha 1/T$

3) V/T = a constant

4) $V_1 T_1 = V_2 T_2$

5) $V_1/T_1 = V_2/T_2$

Select the wrong ones.

A) 1) and 3) B) 2) and 4)

C) 3) and 4) D) 2) and 3)

20. The graph related to volume and temperature of fixed mass of gas at contant pressure always be a .....

A) Parabola B) Staight line

C) Curved line D) Ellipse

21. Two statements related to the temperature and Kinetic energies of gas molecules are given below.

Statement 1 : Gas molecules that are constantly moving possess Kinetic energy.

Statement 2 : When temperature increases, Kinetic energy of gas molecules also increases.

A) Statement 1 is correct and 2 is incorrect

B) Both statements are correct

C) Statement 1 is incorrect and 2 is correct

D) Both the statements are incorrect.

22. Which one of the following is combined gas equation?

A) $\frac{P_1T_1}{V_1} = \frac{P_2T_2}{V_2}$

B) $P_1 V_1 T_1 = P_2 V_2 T_2$

C) $\frac{P_1}{V_1T_1} = \frac{P_2}{V_2T_2}$

D) $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

23. Which among the following substances have same mass:

1) 4 mol Carbon atoms

2) 2 mol Nitrogen atoms

3) 3 mol Sulpher atoms

4) 3 mol Oxigen atoms

(Atomic mass : C = 12, N = 14, S = 32, O = 16)

A) 1) and 2)

B) 3) and 4)

C) 1) and 4)

D) 2) and 3)

24. What is the volume of 2 mol CO_2 at STP?

A) 22.4 L

B) 44.8 L

C) 11.2 L

D) 2 L

25. The measure of average Kinetic energy of molecules in a substance is its .....

A) Mass B) Volume

C) Pressure D) Temperature

26. Number of Oxygen atoms present in 128g Oxygen atom (atomic mass : 16) is ..... $X N_A$.

A) 8 B) 18

C) 128 D) 32

27. During the constant and random motion of gas molecules, they collide with the walls of the container they are in. The gas experiences a characteristic property as a result of this collision is called .....

A) Mass B) Temperature

C) Gasseous Pressure D) Volume

28. If you collect air in a syringe, close the nozzle and press the piston of the syringe, then,

A) Pressure decreases and volume increases

B) Pressure and volume increases

C) Pressure increases and volume decreases

D) Pressure and volume decrease

29. At constant pressure, volume of fixed mass of gas is directly proportional to temperature in Kelvin scale. Which law is explained here?

A) Boyle's law B) Charles law

C) Avogadro's law D) Joule's law

30. Which among the following quantity is one mole.

(Atomic mass : C = 12, N = 14, H = 1, 0 = 16).

A) 224L NH_3 at STP

B) 90g H_2O

C) 16g CH_4

D) 3 x 6.022 x 10^{23} molecules of a substance

31. Choose the wrong statements

a) Any gas containing Avogadro's number of molecules will have a volume of 22.4L at STP.

b) One 'Gram atomic mass' of any element contains one mole of atoms.

c) If we take one molar mass of any substance, their masses will be same.

d) If two gases at the same temperature and pressure have the same volume, the gas with the higher molecular mass will contain more molecules.

A) a) and b) B) c) and d)

C) a) and c) D) b) and d)

32. What is the reason why gas molecules do not lose energy as a result of their collisions.

A) Volume of a gas molecule is negligible as compared to total volume of gas molecules.

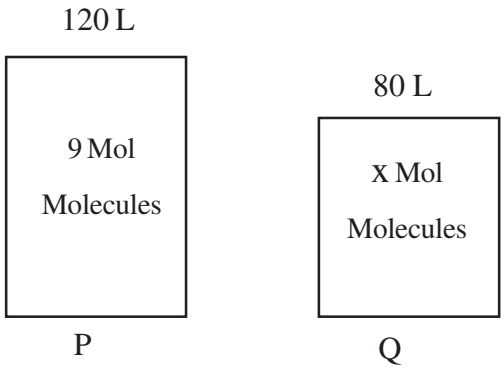
B) Collisions of gas molecules are perfectly elastic

C) Gas molecules move randomly

D) Gas molecules constantly move in all directions.

II Very Short Answer type Questions (2 mark)

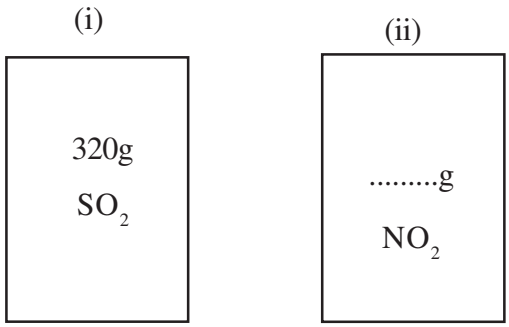
1. Consider the diagram of samples of two different gases, P, Q located at the same temperature and pressure.



a) Calculate the number of moles of particles in the gas Q.

b) Which gas law is explained here?

2. Two gas samples kept at STP containing the same volume are given.



a) Calculate the number of moles of NO_2 in sample (ii)?

b) Calculate the mass of NO_2?

3. a) Caculate the molecular mass of Sulphuric acid (H_2SO_4).

b) Calculate the mass of 1.5 mol CO_2.

(Atomic mass: H=1, S=32, O=16, C=12)

4. 128g of Oxygen gas is stored in a cylinder

(Atomic mass : 16)

a) How many moles of Oxygen molecules does it contains?

b) What will be its volume if it is completely transferred to STP?

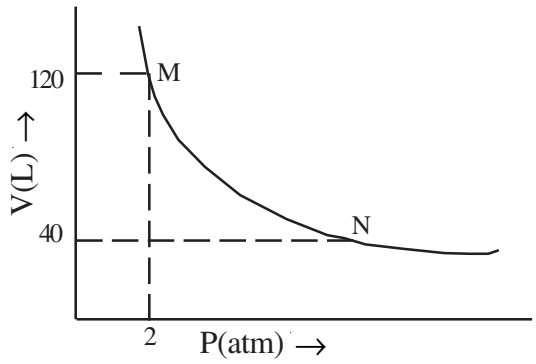
5. The volume and temperature of a fixed mass of gas at constant pressure are listed in various cases.

Cases	V(L)	T(K)
1	28	x
2	60	300
3	y	265

i) What is the temperature (x) in first case?

ii) Calculate the volume (y) of gas in case two?

6. The relationship between pressure and volume when the temperature of a given gas is constant is depicted in the graph.



a) This graph is based on which gas law?

b) What will be the pressure corresponding to the point 'N'?

7. If the volume of a gas at a temperature of 27^0C and a pressure of 5 atm is 600 ml, What is the pressure required to convert the volume of the same gas to 2L at a temperature of 400K.

8. Gas laws describes the behavior of gases.

a) An inflamated balloon burst when exposed to the sun. Give reason?

b) Write the mathematical expression of Boyle's law?

9. Write the difference between the concepts in each of the following.

   1) Molar mass and molar volume

   2) Gram atomic mass and gram molecular mass.

10. The combustion equation for Methane gas is given below.

   $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + Heat$

   How many grams of CO_2 gas will be produced when 5 moles of CH_4 gas are competely burned.

11. Write the ideal gas equation and explain the terms?

12. Calculate the total number of hydrogen atoms present in 224L methane gas at STP.

III Short Answer type Questions (3 mark)

1. If we take the same number of molecules of Carbondioxide gas as the molecules in 36 gram of water.

   a) What will be its mass?

   b) What is the total number of Carbon atoms in this gas?

   c) What will be the volume of the gas if it is at STP?

2. Consider 5 moles of amonia (NH_3) gas at the STP.

   a) What is the total number of molecules contained in it?

   b) Calculate the mass of this amount of ammonia.

   c) What will be the volume of this gas?

3. From the given compounds, find only those are in the gaseous state at normal temperature? (6x½)

1) Sodium Chloride
2) Carbon dioxide
3) Ammonia
4) Ammonium Chloride
5) Glucose
6) Methane
7) Sulphur dioxide
8) Carbon monoxide
9) Nitric acid
10) Nitrogen dioxide

4. i) Explain the reason for the following observations.

a) Bubbles coming from the bottom of the aquarium increases in size as they reach the top.

b) Bottles filled with liquid ammonia are opened after being placed in cold water for a short time.

ii) Write which gas laws are responsible for explaining each of the above situations.

5. Define each of the following.

1) 1 mole
2) Absolute Zero
3) Unified mass
4) Ideal gases

6. The equation for the chemical reaction in which Sulphuric acid waste is neutralized with lime is given.

$$H_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2H_2O$$

Calculate the mass of Sulphuric acid that can be neutralized with 1110 grams of lime $[Ca(OH)_2]$.

7. a) If one mole each of SO_2 gas and NO_2 gas are taken, which one will have the greater mass?

   b) If we take equal masses of ammonia and Carbon dioxide, which one will have more molecules? Why?

8. Calculate the number of Oxygen molecules required to completely burn methane, which has the same mass as 112L of SO_2 gas located in STP.

   (Atomic mass : S=32, O=16, C=12, H=1,
   Equation $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$).

9. Classify the given substances into two moles and three moles based on their number of molecules.

   (atomic mass : H=1, O=16, C=12, Ca=40, N=14)

   1) 36g H_2O

   2) 132g CO_2

   3) 300g $CaCO_3$

   4) $6.022 \times 10^{23} \times 3$ numbers Nitrogen molecules.

   5) 44.8 L Chlorin gas at STP.

10. How many moles of Carbondioxide are required to equalise the mass of 110 moles of Sulphurdioxide?

    (atomic mass : S=32, O=16, C=12)

Chapter 4

GAS LAWS AND MOLE CONCEPT

ANSWER KEY

Objective type (1 mark)

1.	C	11.	C	21.	B	31.	B
2.	B	12.	C	22.	D	32.	B
3.	B	13.	C	23.	C		
4.	B	14.	B	24.	B		
5.	D	15.	B	25.	D		
6.	D	16.	A	26.	A		
7.	C	17.	C	27.	C		
8.	C	18.	B	28.	C		
9.	C	19.	B	29.	B		
10.	B	20.	B	30.	C		

Very short answer (2 marks)

1. a) $V \alpha n \Rightarrow \frac{V_1}{n_1} = \frac{V_2}{n_2}$

$\Rightarrow \frac{120L}{9\ mol} = \frac{80L}{x}$

$\Rightarrow x = \frac{80L \times 9\ mol}{120L}$

= 6 mol

$V_1 = 120L \quad V_2 = 80L$

$n_1 = 9\ mol \quad n_2 = x$

b) Avagadros law

2. a) According to Avagadro's law

Number of moles of SO_2 = Number of moles of NO_2

Number of moles of $SO_2 = \frac{320g}{64g}$ = 5 mol

$\therefore$ Number of moles of NO_2 = 5 mol

b) Mass of 5 mol NO_2 = 5 x Molar Mass

= 5 x 46g

= 230g

3. a) (2x1) + 32 + (4x16)

= 98

b) Mass = n x Molar Mass

= 1.5 x 44g

= 66g

4. a) Number of moles = $\frac{Given\ Mass}{Molar\ Mass}$

$= \frac{128g}{32g}$

= 4 mol

b) at STP, 1 mol = 22.4L

$\therefore$ 4 mol = 4 x 22.4L

= 89.6L

5. a) Case (2) $\frac{V}{T} = \frac{60}{300} = \frac{1}{5}$ (Charle's Law)

Case (1) $\frac{28}{x} = \frac{1}{5}$; $x = 28x5$

= 140K

b) Case (3) $\frac{y}{265} = \frac{1}{5}$

$y = \frac{265}{5} = 53L$

6. a) Boyle's Law

b) PV = a constant

Point M $\Rightarrow$ 2 atm x 120L

= 240L atm

$\therefore$ at Point N $\Rightarrow$ P x 40L = 240L atm

P = 240L atm/40L

= 6 atm

7. Combined gas equation, $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$P_1 = 5$ atm

$V_1 = 600ML$

$= 0.6L$ | $P_2 = ?$

$T_1 = 27^0C$ | $V_2 = 2L$

$= 27+273$ | $T_2 = 400K$

300K

(1) $\Rightarrow P_2 = \frac{P_1V_1T_2}{T_1V_2}$

$= \frac{5\ atm\ x\ 0.6L\ x\ 400K}{300K\ x\ 2L}$ **= 2 atm**

8. a) According to Charle's law, at constant pressure as temperature increases, Volume also increases.

b) $P_1V_1 = P_2V_2$ (or a $V\alpha\frac{1}{P}$, $PV = a$ constant)

9. a) Molar Mass - Mass of one mole of a substance in gram

Molar Volume - Voume of one mole of a gas at a specified temperature and pressure)

b) GAM - Mass of one mole atom in gram

GMM - Mass of one mole molecule in gram.

10. As per balanced equation,

One mole CH_4 produces one mole CO_2

$\therefore$ 5 Mol CH_4 produces 5 mol CO_2

$\therefore$ amount of CO_2 = 5 x GMM of CO_2

= 5 x 44g **= 220g**

11. PV = nRT

where,

P = Pressure

V = Volume

n = number of moles

R = Universal gas constant

T = Temperature in Kelvin

12. Methane $\Rightarrow CH_4$

Number of moles of $CH_4 = \frac{224L}{22.4L}$ **= 10 mol**

One methane contains 4 H atoms

One mole methain contains 4 x 6.022 x 10^{23} H atom

$\therefore$ Number of H atoms in 10 mol CH_4 **= 10 x 4 x 6.022 x 10^{23}**

Short Answer (3 marks)

1. a) $36g\ H_2O \Rightarrow \frac{36}{18} = 2\ mol$

$\therefore$ No. of moles of CO_2 = 2 mol

Mass of CO_2 = 2 x 44g **= 88g**

b) Number of C atoms = 2 mol

= 2x6.022x10^{23}

C) Volume of 2 mol CO_2 at STP = 2x22.4L **= 44.8L**

2. a) $5 \times 6.022 \times 10^{23}$

   b) $5 \times 17 = 85g$

   c) $5 \times 22.4 = 112L$

3. CO_2, NH_3, CH_4, SO_2, CO, NO_2

4.i) a) While moving upwards, pressure decreases and volume increases (Boyle's law)

   b) If the NH_3 contained bottle kept in cold water, NH_3 exists in the liquid state. Otherwise it converts into gaseous from (Charle's law $V \alpha T$).

ii) a) Boyle's law

   b) Charle's law

5. 1) The amount of substance that contains 6.022×10^{23} number of particles.

   2) The lowest imaginary temperature that a gas is to be able to reach is called absolute zero.

   3) It is the mass of $\frac{1}{12^{th}}$ mass of a C-12 atom.

   4) The gas which obey ideal gas equation (PV = nRT) at all conditions of temperature and pressure.

6. Number of moles of 1110g lime = $\frac{1110g}{74g}$ **= 15mol**

   Based on the balanced chemical equation,

   one mole H_2SO_4 reacts with one mol $Ca(OH)_2$

   Number of moles of H_2SO_4 required for the neutralisation of 15 mol lime

   = 15 mol

   = 15 x 98g

   = 1470g

7. a) 1 mol SO_2 = 64g (Molar Mass of SO_2)

   1 mol NO_2 = 46g (Molar Mass of NO_2)

   1 mol SO_2

   b) Let 'x' be the mass of NH_3 & CO_2

No.of moles of $NH_3 = \frac{x}{17}$

No.of moles of $CO_2 = \frac{x}{44}$

$\frac{x}{17} > \frac{x}{44} \Rightarrow$ NH_3 contains more number of molecules

$\Rightarrow$ If mass remains same, as molecular mass decreases number of moles increases. Hence number of molecules increaes.

8. $112L\ SO_2 = \frac{112L}{22.4L}$

= 5 mol

= 5 x 64g

= 320g

$\therefore$ mass of CH_4 = 320g

Balanced equation $\Rightarrow CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Number of moles of $CH_4 = \frac{320}{16} = 20\ mol$

1 mol CH_4 reacts with 2 mol O_2

$\therefore$ 20 mol CH_4 reacts with 40 mol O_2

ie = **40 x 6.022 x 10^{23} O_2 molecules**

9. 1) 36g $H_2O \Rightarrow \frac{36}{18}$ = 2 mol

2) 132g $CO_2 \Rightarrow \frac{132}{44}$ = 3 mol

3) 300 g $CaCO_3 \Rightarrow \frac{300}{100}$ = 3 mol

4) 3 mol

5) 2 mol

2 mol $\Rightarrow$ 1 & 5
3 mol $\Rightarrow$ 2, 3 & 4

10. 110 mol SO_2 = 110 x 64g
= 7040g

$\therefore$ Number of moles CO_2 in 7040g = $\frac{7040g}{44g}$

= 160 mol

Chapter 5

ELECTROCHEMISTRY

I Objective type, MCQ (1 mark)

1. When electric current passes through acidified water, which gases are liberated at the electrodes?

   a) Hydrogen at the Cathode and Oxygen at the Anode
   b) Hydrogen at the Anode and Oxygen at the Cathode
   c) Hydrogen at the Anode and Chlorine at the Cathode
   d) No gases liberated

2. During the electrolysis of Copper Sulphate solution using Copper electrode, the anode:

   a) Gains mass
   b) Loses mass
   c) Remains unchanged
   d) Either gains or losses mass

3. The electrolyte used in electroplating to coat copper on an iron object is,

   a) Dil HCl
   b) Zinc sulphate solution
   c) Copper Sulphate solution
   d) Sodium Chloride solution

4. The colour of the solution fades during the electrolysis os $CuSO_4$ solution, because in solution

   a) Cu^{2+} ions increases
   b) Cu^{2+} ions decreases
   c) SO_4^{2-} ions increases
   d) Water evaporates

5. The electrode at which Oxidation takes place is,

   a) Cathode
   b) Anode
   c) Both the electrodes
   d) None of these

6. Aqueous NaCl on electrolysis produces hydrogen at

a) Cathode b) Anode

c) Both the electrodes d) None of these

7. Electrolysis of water requires

a) Pure water b) Acidified water

c) Distilled water d) None of the above

8. Consider the following galvanic cells

i) Zn - Cu galvanic cell

ii) Mg - Cu galvanic cell

Which has the higher value of voltage.

a) Zn - Cu b) Mg - Cu

c) Same voltage d) No voltage

9. Statement 1 : $CuSO_4$ solution cannot be stored in Zn pot.

Statement 2 : Reactivity of Zn is more than that of Cu.

a) Both the statements are correct

b) Statement 1 is correct & statement 2 is wrong

c) Statement 2 is correct & statement 1 is wrong

d) Both the statements are wrong

10. Which one of the following is not a displacement reaction?

a) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

b) $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2\,Ag$

c) $CaCO_3 \rightarrow CaO + CO_2$

d) $Mg + Cu\,SO_4 \rightarrow MgSO_4 + Cu$

11. Which of the following metals cannot displace hydrogen from dilute HCl?

a) Mg and Zn b) Al and Mg

c) Au and Al d) Au and Ag

12. The compound formed during the electrolysis of aqueous sodium chloride solution is?

a) NaOH b) H_2O

c) KOH d) Both (a) and (b)

13. Button cells are

a) Primary cells b) Secondary cells

c) Fuel cells d) Galvanic cells

14. Power sources used in electric vehicles is

a) Lithium ion battery b) Nickel - Cadmium cell

c) Dry Cell d) Fuel Cells

15. In which of the following vessel silver nitrate solution can be stored safely?

a) Copper b) Aluminium

c) Iron d) None of the above

II Very Short Answer type Questions (2 mark)

16. Molten Sodium Chloride and aqueous Sodium Chloride solution are electrolysed separatly.

a) Which substance is obtained at the anode in both cases?

b) What is the energy change taking place in an electrolytic cell?

17. Gold is coated over a copper spoon by electrolysis

a) Name this process?

b) Which electrolyte used here?

18. There are two types of electrochemical cells.

a) What are they?

b) Write the difference between them.

19. Draw the diagram of Mg-Ag galvanic cell and mark the direction of current?

20. Some metals are given below.

Cu, Al, Mg, Fe, Sn

a) Which metal among these react vigorously with dilute HCl?

b) Which gas is liberated by this reaction?

21. The order of reactivity of some elements are given below.

Cu < H < Ni < Zn < Na

a) Which element loses its luster quickly on exposure to air?

b) Which is the element that cannot displace hydrogen from dil HCl?

22. Dry cell become dead, when the electrical energy produced by chemical reaction is used up.

a) To which category dry cells belongs to?
(Primary cell/Secondary cell)

b) Write its another demerits?

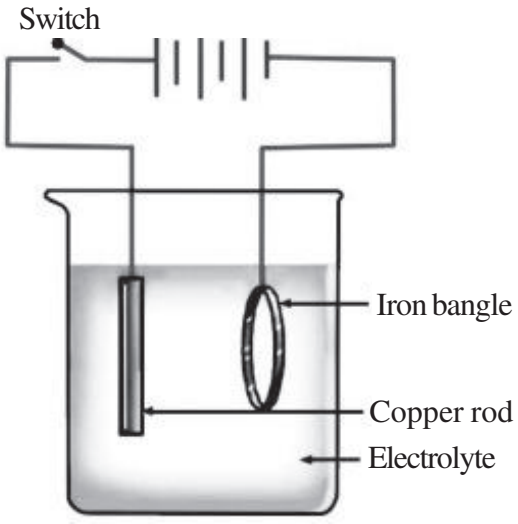
23. Different types of cells are given below.
Classify them suitably and prepare a table.

Lithium ion battery, Button cell Nickel - Cadmium cell, Dry cell

24. a) The energy change taking place in the fuel cell is?

b) Write anyone advantage of fuel cell?

25. Analyse the picture and answer the following questions.

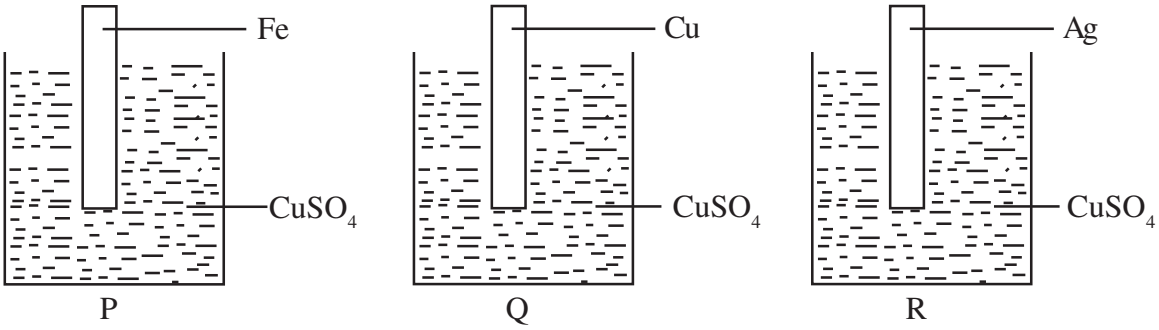


a) Which solution is used as the electrolyte?

b) Write the chemical equation of the reaction taking place at the anode?

III Short Answer type Questions (3 mark)

26. Analyse the picture and answer the following questions.



a) In which beaker does colour change occur?

b) Write the chemical equation of the reaction that causes colour change?

c) Different galvanic cells are constructed using the metals Zn, Ag, Mg and Al which metal acts only as Cathode?

27. Zinc rod is dipped in Copper Sulphate solution.

Observation 1 : Copper is deposited on Zinc

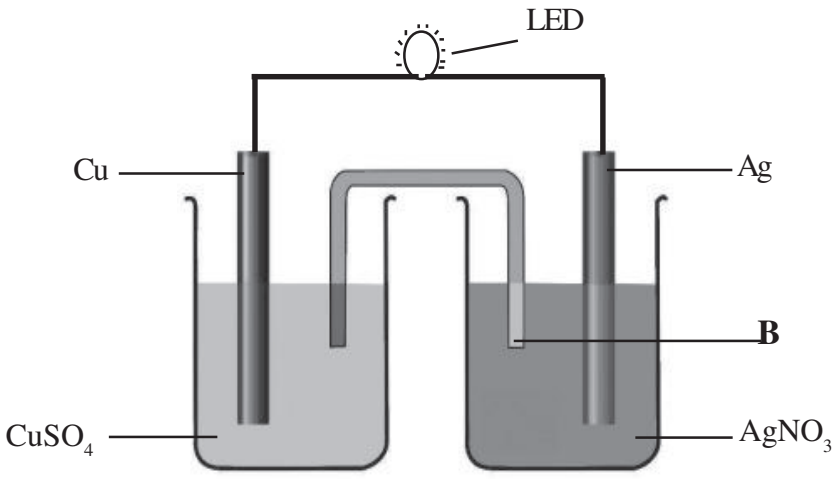
Observation 2 : The intensity of blue colour of Copper Sulphate solution decreases.

a) Which metal is less reactive?

b) Name the reaction taking place here?

c) Write the chemical equation for the above observation.

28.

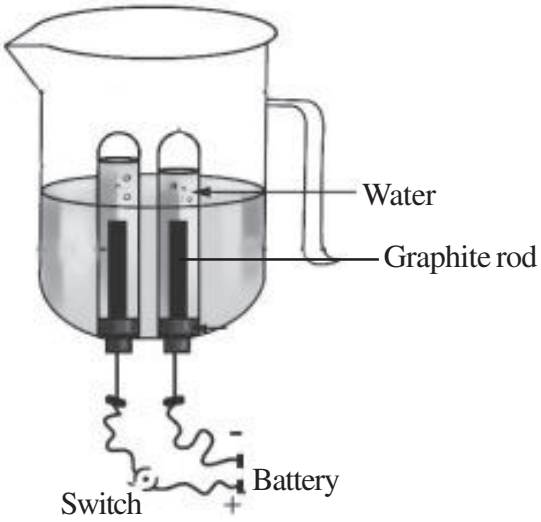


a) Identify the cell?

b) What does 'B' represent?

c) How does 'B' prepared?

29. A picture of electrolysis of acidified water is given below.



a) Which type of cell is represented in this figure?

b) Write the chemical equation of the cell.

c) Name the electrode in which oxidation takes place.

30. Anode : $Cu \rightarrow Cu^{2+} + 2e^{-}$

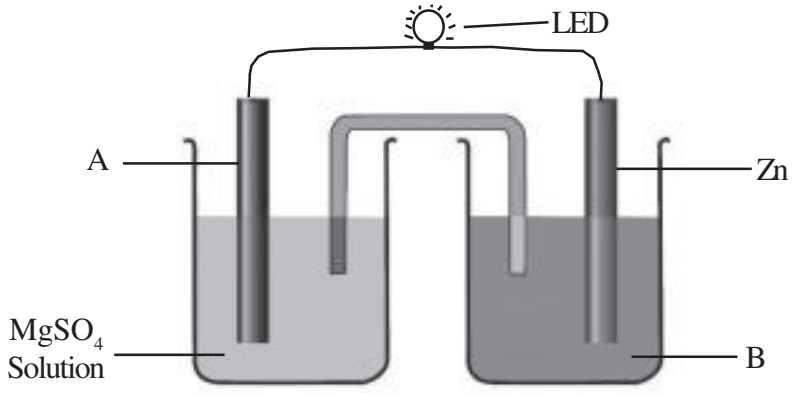
Cathode : $Cu^{2+} + 2e^{-} \rightarrow Cu$

a) Based on above reaction draw a diagram of electroplating of Iron bangle.

b) Write any one use of (merit) electroplating?

IV Short Answer type Question (4 mark)

31. The given diagram represents Galvanic cell.



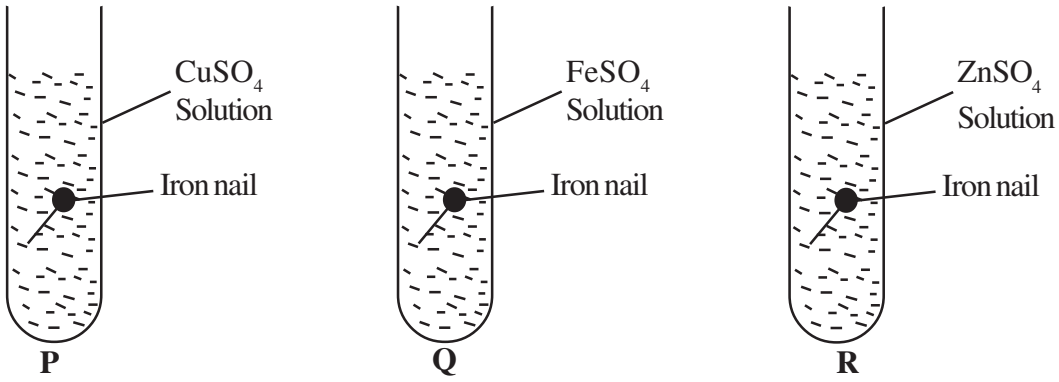
a) Identify A and B?

b) As the cell operates the mass of which electrode decreases?

c) Write the direction of flow of electrons.

d) Write the cell reaction.

32. Analyse the figure and answer the following questions.



a) In which test tube the iron nail undergoes a colour change?

b) Write the reason for the colour change?

c) Write the name of the reaction taking place here?

d) Write the chemical equation of this reaction.

33. There are three pairs of electrodes

i) Al - Cu ii) Zn - Cu iii) Cu - Ag

a) Which pair of electrodes has the higher value of voltage?

b) Write the reason for answer?

c) Write any two appication of reactivity series?

34. Consider the displacement reaction and answer the questions.

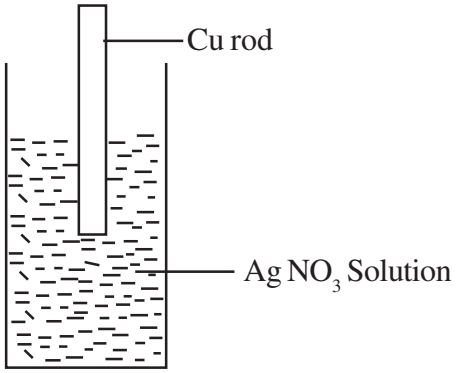
$$Mg + \overset{2+}{C}u\overset{2-}{S}O_4 \rightarrow \overset{2+}{M}g\overset{2-}{S}O_4 + Cu$$

a) Which metal gets oxidised here?

b) Copper ion undergoes which type of reaction?

c) Name the oxidising and reducing agents in this reaction?

35.



Based on above diagram answer the questions.

a) What colour change do you observe in this reaction?

b) Which ion is responsible for the change of colour in this solution?

c) Which metal is reduced here?

d) Write the equation for Oxidation?

Chapter 5

ELECTROCHEMISTRY

Answers

Objective type (1 mark)

1. a)
2. b)
3. c)
4. b)
5. b)
6. a)
7. b)
8. b)
9. a)
10. c)
11. d)
12. a)
13. a)
14. a)
15. d)

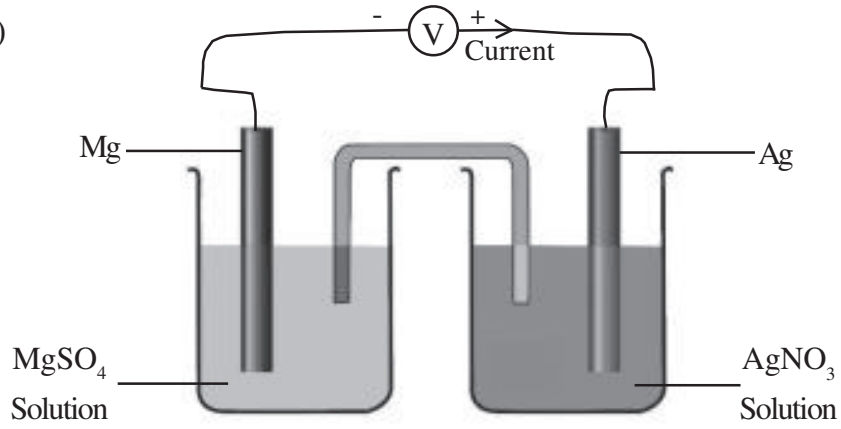
Very Short Answer (2 mark)

16. a) Chlorine
    b) Electrical energy $\rightarrow$ Chemical energy

17. a) Electroplating
    b) Mixture of gold cyanide and sodium cyanide solution

18. a) Galvanic cell and Electrochemical cell
    b) Galvanic cells convert chemical energy into electrical energy
    Electrochemical cell convert electrical energy into chemical energy

19. a)



20. a) Mg
    b) Hydrogen

21. a) Na
    b) Cu

22. a) Primary cell
    b) Primary cells cannot be recharged and reused

23.

Primary Cell	Secondary Cell
Dry Cell	Lithium ion battery
Button cell	Nickel Cadmium Cell

24. a) Chemical energy into electrical energy
    b) They produce electricity with high efficiency

25. a) Copper Sulphate Solution
    b) $Cu \rightarrow Cu^{2+} + 2e^-$

Short Answer (3 mark)

26. a) P
    b) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
    c) Ag

27. a) Cu

b) Displacement reaction

c) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

28. a) Galvanic Cell

b) Salt Bridge

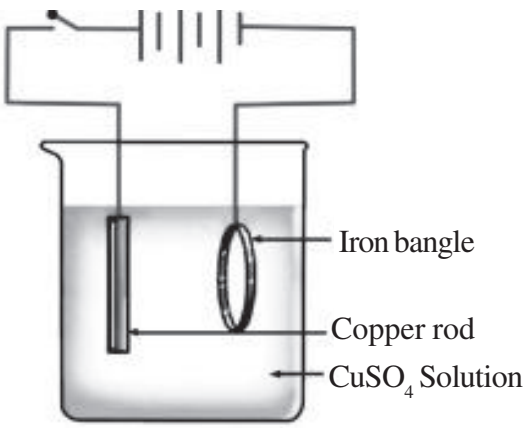
c) Take a piece of rolled filter paper and add a little Potassium Chloride into it and then sprinkle water to moisten it. This filter paper can be folded into 'U' shape and immerse in two beakes.

29. a) Electrolytic cell

b) $2H_2O \rightarrow 2H_2 + O_2$

c) Anode

30. a)



b) Enhances the beauty of the metal and prevents its corrosion.

Short Answer (4 mark)

31. a) A - mg

B - $ZnSO_4$

b) Anode (Mg)

c) $Mg \rightarrow Zn$

d) $Mg_{(s)} + \overset{2+}{Zn}_{(aq)} \rightarrow \overset{2+}{Mg}_{(aq)} + Zn_{(s)}$

32. a) P

b) Iron is more reactive than Cu. Therefore Iron displaces Cu from its salt solution.

c) Displacement reaction

d) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

33. a) i) Al - Cu

b) Oxidation tendency of Al is more than Cu and Zn

c) i) To identify which metal is being displaced

ii) To identify oxidising and reducing agent

34. a) Mg

b) Reduction

c) Cu^{2+} is the Oxidising agent ($CuSO_4$)

Mg is the reducing agent

35. a) Solution turns blue

b) Cu^{2+}

c) Ag

d) $Cu \rightarrow Cu^{2+} + 2e^-$

Chapter 6

METALS

I Objective type, MCQ (1 mark)

1. Which of the following is an ore of aluminium?
   a) Haematite b) Bauxite
   c) Cuprite d) Calamine

2. Calamine is the ore of which metal?
   a) Aluminium b) Iron
   c) Copper d) Zinc

3. Which of the following ore is concentrated by froth flotation process?
   a) Zinc blende b) Cuprite
   c) Haematite d) Tinstone

4. Statement 1 : Calcination refers to heating the concentrated ore in limited quantities or absence of air at a temperature below the melting point.
   Statement 2 : Calamine, an ore of zinc, is converted to oxide through calcination
   a) Statement 1 is true but 2 is wrong
   b) Statement 1 is wrong but 2 is true
   c) These two statements are true
   d) These two statements are wrong

5. Which concentration method seperates the gangue iron tungstate from the ore of tin, tinstone
   a) Levigation b) Froth floatation
   c) Magnetic seperation d) Leaching

6. The impurities contained in the ore is generally called .....
   a) Gangue b) Flux
   c) Slag d) None of these

7. Which of the following is the method of concentration of bauxite, the ore of aluminium?

a) Levigation b) Froth floatation
c) Magnetic seperation d) Leaching

8. Which of the following method is used to convert calcium carbonate into calcium oxide?

a) Calcination b) Roasting
c) Froth floatation d) Reduction

9. Heating the concentrated ore at a temperature below the melting point in the presence of air is known as .....

a) Calcination b) Roasting
c) Reduction d) None of these

10. Metals such as tin and lead having low melting point can be refined by .....

a) Liquation b) Distillation
c) Electrolytic refining d) Evaporation

11. Statement 1 : When iron is produced in a blast furnace, Calcium Carbonate, Coke is fed into the furnace along with the ore.
Reason : Coke is added to the furnace to produce carbomonoxide gas, which reduce the iron ore.

a) Statement is true but the reason is false
b) Both statement and reason are true
c) Both statement and reason are wrong
d) None of these

12. Which of the following is not a mineral of aluminium?

a) Clay b) Bauxite
c) Cryolite d) Cassiterite

13. When iron is produced using blast furnace, which compound is used as a reducing agent?

a) Electricity b) Limestone

c) Carbon monoxide d) Coke

14. Which of these is used as a flux, when iron is produced using blast furnace?

a) Calcium carbonate b) Calcium silicate

c) Calcium Oxide d) None of these

15. Statement 1 : In the production of aluminium electricity is used as a reducing agent to seperate aluminium metal from its ore.

Statement 2 : The separate highly reactive metals from their ores, powerful reducing agents are required. Therefore electricity is used.

a) Both statements are correct. Statement 2 is the correct explanation of statement 1

b) Statement 1 is correct, Statement 2 is not the correct explanation of statement 1

c) Both statements are wrong

d) None of these

16. Check the following statement and reason regarding the electrolysis of Aluminum Oxide and select the correct answer.

Statement : The mixture alumina and molten cryolite is used as electrolyte in the production of aluminium.

Reason : Cryolite is added to reduce the melting point and to increase the electrical conductivity.

a) The statement is correct, but the reason is not correct

b) The statement is not correct, but the reason is correct

c) The statement and reason are correct. The correct explanation of the statement is given as the reason.

d) The statement and reason are correct. The reason given is not a correct explanation of the statement.

17. High purity Copper metal is obtained by

a) Reduction using coke

b) Electrolytic refining

c) Liquation

d) Reduction using Hydrogen

18. Observe the following statements regarding Silicon Steel.

i) The constituent elements are Iron, Manganese and Carbon

ii) It is known as electrical steel

iii) It is used in the core of transformer

iv) It is used for the manufacture of railway tracks

a) i) and ii) are correct b) ii) and iii) are correct

c) iii) and iv) are correct d) i) and iv) are correct

19. Observe the table and choose the correct answer

Alloy	Constituent Elements
a) Alnico	i) Cu, Zn
b) Manganese Steel	ii) Fe, Cr, Ni, C
c) Brass	iii) Fe, Mn, C
d) Stainless Steel	iv) Fe, Al, Ni, Co

	(a)	(b)	(c)	(d)
A)	(iv)	(iii)	(i)	(ii)
B)	(iii)	(ii)	(i)	(iv)
C)	(i)	(ii)	(iii)	(iv)
D)	(ii)	(iv)	(iii)	(i)

20. Which metal is produced through Hall-Heroult process?

a) Iron b) Zinc

c) Copper d) Aluminium

21. Which metal will corrode when two metals are in contact in the presence of moisture in the air? Choose the correct answer.

i) The more reactive metal
ii) The less reactive metal
iii) The metal acting as anode
iv) The metal acting as cathode

a) i) and iv) b) ii) and iii)
c) i) and iii) d) ii) and iv)

22. The Iron containing 0.2% to 0.6% Carbon is known as .....

a) Mild Steel b) Medium steel
c) High carbon steel d) None of these

23. The features of steel used in the manufacture of agricultural tool are given below. Which of the following are correct?

i) The amount of Carbon is 0.05% - 0.2%
ii) It is medium steel
iii) It has very high elasticity and hardness
iv) Easy to draw into wires and hammer into plates

a) i) and ii) b) i) and iii)
c) i) and iv) d) ii) and iii)

24. Which of the following is not a Sulphide ore?

a) Copper pyrites b) Zinc blende
c) Calamine d) Copper glance

II Very Short Answer type Questions (2 mark)

1. The ores of some metals are given.

Haematite (Fe_2O_3), Magnatite (Fe_3O_4), Copper pyrites ($CuFeS_2$)

a) Which of these ore is concentrated by frothfloatation process?
b) What is the reason for choosing this method for concentration?

2. The chemical reaction that converts concentrated ore into its Oxide is given below

$$2\,ZnS_{(s)} + 3O_{2(g)} \xrightarrow{Heat} 2\,ZnO_{(s)} + 2SO_{2(g)}$$

a) Write the name of this method?

b) Describe this process.

3. The chemical reactivity of some metals are given below

Al > Zn > Fe > Au

a) Which of these metal compounds is most stable?

b) Which metal is found in free state in nature?

4. Iron is produced using blast furnace.

a) What is the name of iron obtained from blast furnace?

b) What is the name of the ralatively pure iron obtained by reducing the percentage of impurities?

5. a) Write the equation of the formation of slag, when iron is produced using blast furnace?

b) Write any one of the uses of slag?

6. a) Write the name of any metal that is produced by liquation?

b) What is the reason for choosing this situation?

7. a) Which reducing agents are used to seperate moderately reactive metals (eg: Fe, Zn, Sn) from their oxide ores?

b) What is the special feature of this reducing agent that is being utilized here?

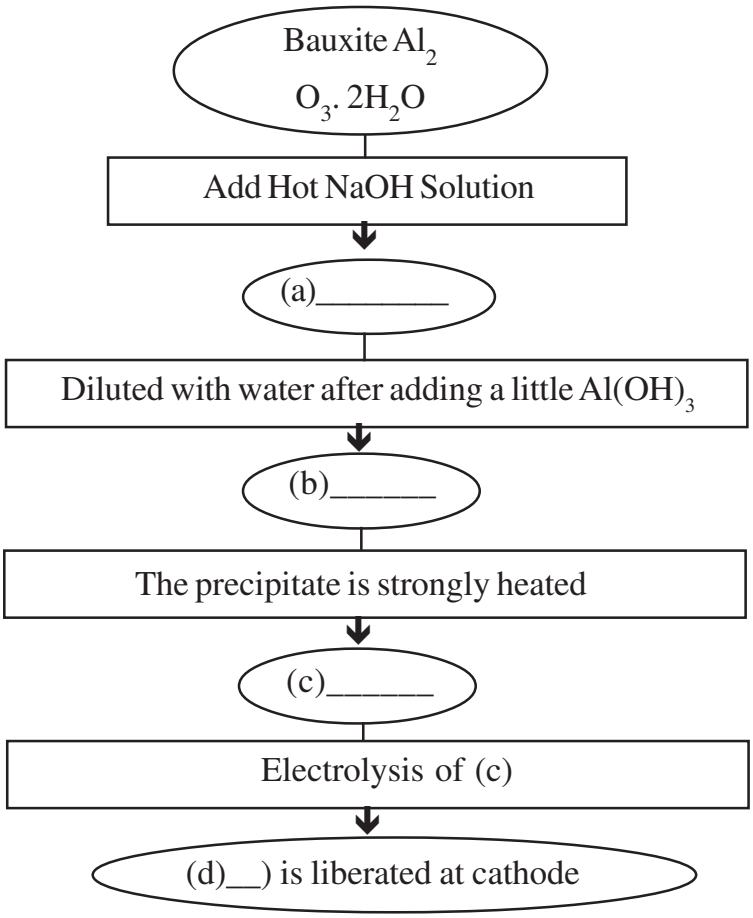
8. a) Write the chemical formula of the Carbonate Ore of Zinc?

b) Which technique is used to convert this ore into its Oxides?

9. a) Write the name of any metal that is found almost in free state in nature?

b) Which solvent is chosen when leaching is used to concentrate these metals?

10. Fill in the blanks given by flow chart showing the various stages of aluminium manufacturing.



11. a) What is the name of iron obtained from the Blast furnace?
    b) The iron obtained in this way is converted into steel and used for various industrial purpose. What is the need to convert it into steel?

12. What is the need to connect underground iron pipes and storage tanks with Magnesium blocks?

13. Brass and Bronze are the important alloys of a metal.
    a) Identify the metal
    b) Which flux is used in the production of this metal? Why?

14. Duralumin is an important alloy containing Aluminium.
    a) What is an alloy? What is the importance of making it an alloy?
    b) Write the name of another alloy of aluminum?

15. When iron materials are joined to copper piping for electrical purpose, the iron quickly corrodes through contact.
    a) When iron comes into contact with copper, which metal will release electrons? Why?
    b) What is mean by metal corrosion?

III Short Answer type Questions (3 mark)

1. Iron is extracted from ore using a system called blast furnace?
    a) Which iron ore is used here?
    b) What are the raw materials added to the blast furnace?
    c) What are the gangue and flux from these?

2. Complete the table given below related to metal refining.

Process	Characteristics based on	The metals refining by this
Liquation	a)	b) .....
c)	Low boiling point	d) .....
e)	Conduct electricity	f) .....

3. Iron is an industrially important metal
    a) Which ore is used for the industrial production of iron?
    b) Why is limestone added into to the furnace?
    c) Write the chemical fomula of the slag removed from the blast furnace?

4. The iron obtained from the blast furnace contains small amount of carbon and other impurities.
    a) Write the name of the iron obtained from the blast furnace?
    b) The iron containing approximately 2% of carbon?
    c) Relatively pure iron?

5. a) Write two factors that influence the corrosion of metals?
    b) How does the metallic corrosion of iron differ from the metallic corrosion of aluminium?

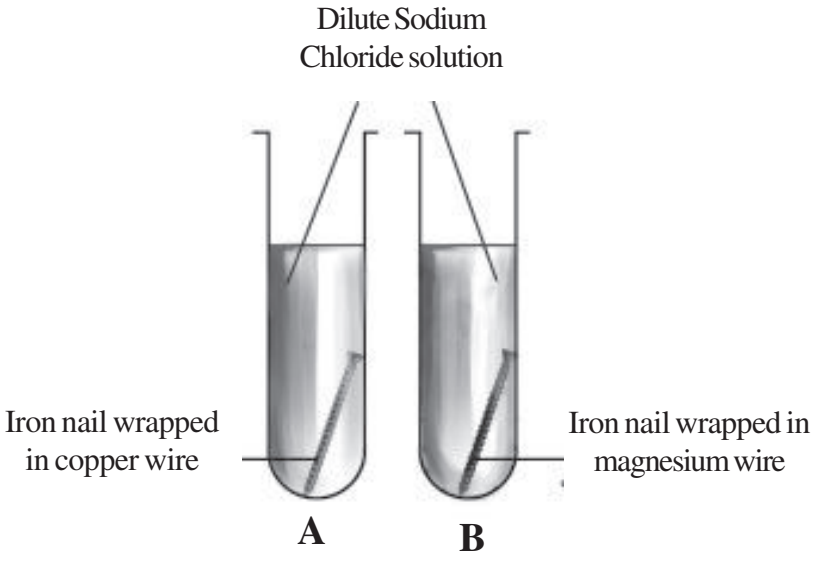
6. Fill in the blanks

Metal	Alloy	Use
(a) _____	Alnico	(b) _____
Zinc	(c) _____	Manufacturing of door handle
(d) _____	Silicon Steel	(e) _____
(f) _____	Bronze	Manufacturing of coins

7. a) How much (percentage) carbon is contained in steel?

   b) Which steel has the highest content of carbon?
   Write one use of it.

8.



Dilute NaCl solution is taken in two test tubes A and B. Place iron nails, one tighty wrapped with Copper wire and the other with Magnesium wire.

a) What will be the result of the observation after a few days?

b) The iron nails in which test tube is protected?

c) What is the name of this method of preventing corrosion of iron?

9. a) Draw the diagram of refining of Copper.
   Label the anode, cathode and electrolyte in the figure.

   b) Write down the chemical equation of the reaction taking place at cathode.

10. Which metal alloy is used for the following uses?

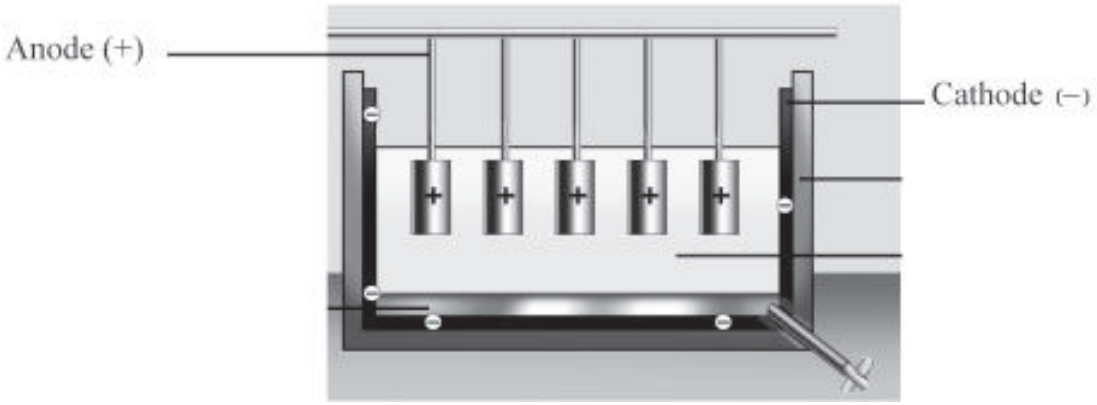
a) Motor and Generator

b) Blades

c) Permanent magnets

d) Coins and Medals

11.



Observe the diagram and answer the following questions.

a) Production of which metal is shown in the diagram?

b) Which substance is used here as electrolyte?

c) Which gas is liberated at the anode?

IV Short Answer type Question (4 mark)

1. Complete the table

Nature of the ore	Concentration method	Example
Density of the ore is more than that of gangue	a)	iron
b)	Magnetic seperation	Tinstone
c)	Froth floatation	Sulphide
Ore is soluble in a suitable solvent	Leaching	d) .....

2. Iron is produced industrially, mainly from haematite

   a) Which substance is used to reduce haematite in iron making? How is it produced in a furnace?

   b) Which gangue is present mainly in haematite? Which compound is used to re move this gangue?

3. Some chemical equations are given below relate to the manufacture of iron.

   i) $CO_2 + C \longrightarrow 2CO$

   ii) $Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$

   iii) $CaCO_3 \longrightarrow CaO + CO_2$

   iv) $CaO + SiO_2 \longrightarrow CaSiO_3$

   a) Which of the compound act as reducing agent?

   b) Which process is responsible for the formation of slag?

   c) Which activity shows the reduction of ore?

   d) Write the name of the iron obtained from blast furnace?

4. Answer the following questions related to the production of Copper.

   a) Which are the main ores of Copper?

   b) What s the method of concentraton of these ores?

   c) Reducing agent is not added to convert Cuprous Sulphide (Cu_2S) into Copper after concentration. Why?

   d) The Copper lumps obtained during the production of Copper are called Blister Copper. Why?

5. Following are two facts related to the manufacture of an industrially important metal.

   i) Its Sulphide ore is concentrated through Froth Floatation Process.

   ii) Due to its low boiling point during metal production, It is immediately cooled to liquid form.

   a) These facts are related to the production of which metal?

b) What is the reason for using Froth Floatation Process for ore concentration?

c) Which process converts this ore into Oxide after concentration?

d) What is the method of purification of this metal?

6. Answer the following questions regarding the electrolysis of aluminium oxide (Alumina)

a) What is used as Anode? What reaction takes place at anode?

b) What is used as Cathode? What reaction takes place at Cathode?

c) In the electrolytic cell, the positively charged rods are replaced periodically. Why?

•

Chapter 6

METALS

ANSWERS

Objective type (1 mark)

1. b)
2. d)
3. a)
4. c)
5. c)
6. a)
7. d)
8. a)
9. b)
10. a)
11. b)
12. d)
13. c)
14. c)
15. a)
16. c)
17. b)
18. b)
19. A)
20. d)
21. c)
22. b)
23. a)
24. c)

Very Short Answer (2 mark)

1. a) $Cu\,Fe\,S_2$
   b) Sulphide ores

2. a) Roasting
   b) Concentrated Ore is heated in a current of air below its melting point.

3. a) Aluminium (Al)
   b) Gold (Au)

4. a) Pig iron
   b) Rought iron

5. a) $CaO_{(s)} + SiO_{2(s)} \longrightarrow CaSiO_{3\,(\ell)}$
   b) Manufacture of cement and road

6. a) Tin (Sn)
      Lead (Pb)
   b) Low melting point

7. a) Coke, Carbon monoxide
   b) Carbon has a greater tendency to react with oxygen at high temperatures, So oxygen is easily removed from the ore.

8. a) $Zn\,CO_3$
   b) Calcination

9. a) Gold/Silver
   b) dil NaCN/dil KCN

10. a) Sodium Aluminate or $NaAlO_2$
    b) Aluminum Hydroxide or $Al(OH)_3$
    c) Alumina or Al_2O_3
    d) Aluminium or Al

11. a) Pig iron
    b) The pig iron obtained from the blast furnace is not suitable either for hammering into various shapes or for drawing into wires.

12. It is a method to prevent corrosion of iron (Cathodic protection). The more reactive metal (Magnesium) acting as anode is destroyed and the metal (Iron) acting as Cathode is protected.

13. a) Copper or Cu
    b) Sand (SiO_2) having acidic nature
    Because the ore containing basic impurities

14. a) Alloys are mixtures of two or more metals. They have high hardness. Also they resist corrosion.
    b) Alnico

15. a) Iron. Iron is highly reactive than Copper.
    So Iron is act as anode and it is Oxidised.
    b) Corrosion of metal is a process in which the metal react with a surrounding

III Short Answer (3 mark)

1. a) Haematite
    b) Haematite, Limestone, Cock
    c) Gangue - Silica
    Flux - Calcium Oxide

2. a) Low Melting point
    b) Sn, Pb
    c) Distillation
    d) Zn, Cd, Hg
    e) Electrolytic refining
    f) Cu, Zn

3. a) Haematite
    b) To remove the gangue sand

c) $CaSiO_3$

4. a) Pig iron
   b) Cast iron
   c) Rought iron

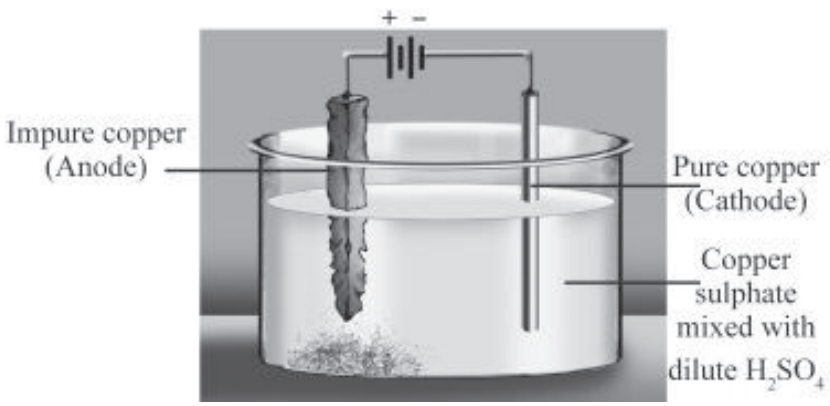
5. a) Nature of the metal, Moisture in which gases or salts are dissolved, contact with other metals. (any 2)
   b) The hydrated Iron Oxide that forms on the surface of Iron is porous and powdery. So this process of corrosion continues until the metal is completely destroyed. But the Oxide coatings on aluminium are thin and nonporous and therefore protect the metal from further corrosion.

6. a) Aluminium (Al)
   b) Making permanent magnets
   c) Brass
   d) Iron
   e) Used in the core of electromagnetic instruments
   f) Copper

7. a) 0.05% to 1.5%
   b) High Carbon Steel
   Used for the manufacture of surgical instruments, springs, knives, drills etc.

8. a) The iron nail in the test tube A undergoes rusting.
   b) Test tube B
   c) Cathodic protection

9. a)



b) $Cu^{2+} + 2e^{-} \rightarrow Cu$

10. a) Silicon Steeel
    b) Stainless Steel
    c) Alnico
    d) Bronze

11. a) Aluminium
    b) Molten Cryolite and Alumina
    c) Oxygen

IV Short Answer (4 mark)

1. a) Levigation or Hydraulic washing
    b) Ore or impurity is magnetic in nature
    c) Ore is less dense than the gangue
    d) Bauxite

2. a) Carbon monoxide

    Coke react with Oxygen and form CO_2

    $C + O_2 \longrightarrow CO_2 +$ heat

    Coke combine with Carbondioxide (CO_2) to form Carbon Monoxide (CO)

    $CO_2 + C + heat \longrightarrow 2CO$

    b) Silica (SiO_2)

    Calcium Oxide (CaO)

3. a) CO
    b) $CaO + SiO_2 \longrightarrow CaSiO_3$
    c) $Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$
    d) Pig iron

4. a) Copper Pyrites or $CuFeS_2$

    Copper glance Or Cu_2S

b) Froth Floatation

c) By heating with flux (SiO_2) Cu_2S is partially converted to Cuprous oxide (Cu_2O). This Cu_2O reduced the remaining Cu_2S to Cu.

d) The SO_2 gas produced during the production of Copper is released when the metal condenses. As a result blisters are formed and they appear as such on the surface of Copper lumps.

5. a) Zinc

b) The Sulphide ore is less dense than the gangue

c) Roasting

d) Distillation

6. a) Carbon rods

$2O^{2-} \rightarrow O_2 + 4e^-$

b) Steel tank with Carbon lining

$Al^{3+} + 3e^- \rightarrow 3Al$

c) Oxygen released at the anode (Carbon rod) Oxidises the carbon rod to oxides of carbon.

As a result the size of the anode rod decreases.

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Chapter 7

SOME COMPOUNDS OF INDUSTRIAL IMPORTANCE

I Objective type, MCQ (1 Mark)

1. Select the wrong statements from the following?

   A) Ammonia is a gas

   B) Ammonia is Denser than air

   C) Ammonia dissolves in water

   D) Ammonia is acidic in nature

   a) A & B b) B & C

   c) C & D d) B & D

2. $\underline{\quad A \quad} + Ca(OH)_2 \rightarrow \underline{\quad B \quad} + CaCl_2 + H_2O$

   $B + H_2O \rightarrow NH_4OH$

   Identify A & B

   a) $A = NH_3$ $B = NH_4Cl$

   b) $A = NH_4Cl$ $B = NH_3$

   c) $A = NaCl$ $B = NaOH$

   d) $A = NH_4Cl$ $B = HNO_3$

3. In what ratio are N_2 & H_2 combined in the industrial production of NH_3

   a) 2:3 b) 3:1

   c) 1:3 d) 1:2

4. Ammonium chloride reacts with Sodium hydroxide solution to form a pungent smelling gas. This gas reacts with hydrochloric acid to form a thick white smoke. Which chemical in this thick white smoke?

   a) NaCl b) KCl

   c) $CaCl_2$ d) NH_4Cl

5. Match the following

A	B
i) Liquor ammonia	P Sodium hydroxide
ii) Liquid ammonia	Q Drying agent
iii) Caustic Soda	R Liquified NH_3
iv) Calcium Oxide	S Concentrated aqueous solution of NH_3

	(i)	(ii)	(iii)	(iv)
a)	R	S	P	Q
b)	S	R	P	Q
c)	P	Q	R	S
d)	S	R	Q	P

6. Statement 1 : Reaction between ammonia and hydrochloric acid is a reversible reaction.

Statement 2 : Products do not become reactants in a reversible reaction

a) Both the statements are true

b) Statement one is false, statement two is true

c) Statement two is false, statement one is true

d) Both the statements are false

7. Select the correct statement

a) In a state of equilibrium, the reactants are completely converted into products.

b) In a state of equilibrium the rates of forward and backward reactions are not equal

c) Chemical equilibrium is a dynamic equilibrium at the molecular level

d) Chemical equilibrium is not possible in a closed system

8. $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ Under what circumstances can the quantity of products be increased in this system at equilibrium?

A) Add more N_2

B) Remove H_2

C) Remove NH_3

D) Add more NH_3

a) A & B b) A &C

c) B & C d) C &D

9. If we increase the concentration of hydrogen in the equilibrium system given below.

$$H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$$

a) Rate of forward reaction increases

b) Rate of backward reaction decreases

c) Rate of backward reaction increases

d) There is no change in the rates of forward and backward reactions

10. The optimum temperature for the industrial production of NH_3 is .....

11. In the equilibrium system given below

$$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$$

Statement 1 : When pressure, increase the rate of forward reaction also increases

Statement 2 : Pressure has no effects

a) Statement 1 is true

b) Statement 1 and 2 are true

c) Statement 2 is true

d) Statement 1 & 2 are false

12. In which of the following reactions does the amount of product increase when pressure is increased.

a) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

b) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

c) $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g}$

d) $2SO_{3(g)} \rightleftharpoons 2SO_{2(g)} + O_{2(g)}$

13. $A_{2(g)} + 3B_{2(g)} \rightleftharpoons 2AB_{3(g)} +$ heat

What change in temperature should be made to increase the amount of product in this reaction?

14. In a reversible reaction, catalysts

a) Increases only the rate of forward reaction

b) Increases only the rate of backward reaction

c) Increases the rate of forward and backward reactions

d) Cannot be predicted

15. In the industrial production of sulphuric acid, the product formed by dissolving SO_3 in concentrated H_2SO_4 is,

a) $H_2S_2O_3$ b) $H_2S_2O_7$

c) H_2SO_3 d) $H_2S_2O_6$

16. When a little H_2SO_4 is added to sugar, a black substance is formed. Which substance is this?

a) H_2 b) S

c) C d) P

17. Statement 1 : Sulphuric acid when diluting, add sulphuric acid little by little to water and stir

Statement 2 : Sulphuric acid is denser than water

a) Statement 1 is true & statement 2 is false

b) Statement 1 is false, statement 2 is true

c) Statement 1 is true & statement 2 is the correct explanation of statement 1

d) Statement 1 is true & statement 2 is not the correct explanation of statement 1

18. What products are obtained when potassium chloride salt is added to the acid obtained by diluting oleum?

a) $KHSO_4$ & HCl b) KNO_3 & HCl

c) $KHSO_4$ & HNO_3 d) None of these

19. $Cu+2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

In this reaction,

a) Copper undergoes reduction & H_2SO_4 undergoes oxidation

b) Copper and H_2SO_4 undergoes oxidation

c) Copper and H_2SO_4 undergoes reduction

d) Copper undergoes oxidation & H_2SO_4 undergoes reduction

20. Some acids and their chemical formula are given.

Which of these are dibasic acid?

a) H_2SO_4 : Sulphuric acid

b) HNO_3 : Nitric acid

c) H_3PO_4 : Phosphoric acid

d) H_2CO_3 : Carbonic acid

A) a & c B) b & d

C) a & d D) c & d

21. Match the following

A	B
1. Ammonia	a) Chlor alkali process
2. Sulphuric acid	b) Haber process
3. Sodium hydroxide	c) Ostwald process
4. Nitric acid	d) Contact process
	e) Hall-Heroult process

	1	2	3	4
A)	b	d	a	c
B)	b	c	d	a
C)	c	d	a	e
D)	d	e	a	c

22. Which of the following salt solution turns blue litmus to red?

a) Na_2CO_3 b) NH_4Cl

c) K_2CO_3 d) CH_3COONa

23. Among the various nutrients required by plants, the following elements belong to which category?

Nitrogen (N), Phosphorous (P), Potassium (K)

a) Natural nutrients

b) Primary nutrients

c) Secondary nutrients

d) Micro nutrients

2 Marks

1. a) When Ammonium Chloride (NH_4Cl) react with Calcium hydroxide [$(Ca(OH)_2$] a bad smell gas is formed. Identify that gas?
   b) Write the Chemical Equation of this reaction?

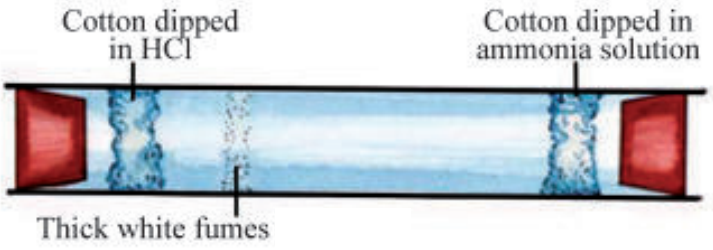
2. a) The drying agent used in the preparation of Ammonia in the laboratory?
   b) What is the use of drying agent here?

3. a) During the preparation of NH_3 in the laboratory, the jar used for collecting ammonia is kept inverted. Why?
   b) How can we identify the basic nature of NH_3?

4. Write the product obtained, when NH_3 is dissolved in water?
   Write the chemical equation of this process?

5. Write the difference between liquid ammonia and liquor ammonia?

6. The reaction between Ammonia and Hydrogen Chloride is given below.



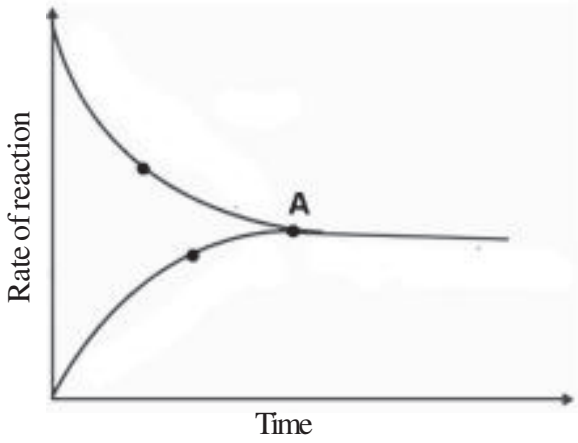
   a) Identify the compound having white dense fume?
   b) Write the chemical equation of this reaction?

7. From the given reversible reactions, write the equation of backward reaction and identify the reactants?

   a) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

   b) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

8.



a) 'A' represents .....

b) Write the two properties of 'A'?

9. a) Write the name of catalyst used for the industrial preparation of NH_3?

   b) How catalyst influences chemical equilibrium?

10. a) Write the name of the product obtained when SO_3 is dissolved in Conc. Sulphuric acid?

    b) Write the Chemical Equation of this process?

11. Why SO_3 is not directly dissolved in water during the industrial preparation of H_2SO_4?

12. Which Chemical property of H_2SO_4 is used in the following situations.

    a) H_2SO_4 is used to remove moisture content during the preparation of HCl.

    b) When Con. H_2SO_4 is added to sugar, it turns black.

13. a) Identify A and B.

    i) $NaCl + H_2SO_4 \rightarrow A + B$

    ii) $KNO_3 + H_2SO_4 \rightarrow A + B$

14. a) $C + H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$

    Identify the oxidising agent in this reaction.

    b) What is the basicity of acid used in the reaction?

15. a) What is Arrhenius concept?

    b) Write the ionic equations, when Ammonium hydroxide is dissolved in water?

16. a) Name the process in which NaOH is industrially prepared?
    b) During the preparation of NaOH, Brine solution is electrolysed. Write the name of this electrolytic cell?

17. a) What are the ions obtained, when NaCl is dissolved in water?
    b) Write any two uses of Sodium hydroxide?

18. a) The reactor used for preparing HCl is known as 'HCl oven'. Why
    b) Write any two uses of HCl?

19. Classify the following salts as acidic, basic and neutral.

    NH_4Cl, Na_2CO_3, KCl, CH_3COONa, $(NH_4)_2SO_4$, NaCl

20. Write any two essential qualities of fertilizers?

Short Answer type Questions (3 mark)

1. The Haber process is the industrial production of NH_3
    a) Check the chemical equation given below and correct it if there is any mistake

    $$N_{2(g)} + 3H_{2(g)} + \text{Heat} \rightleftharpoons[V_2O_5] 2NH_{3(g)}$$

    b) What is the reason for using a relatively high temperature of 450^0C in the industrial production of ammonia?
    c) Write any two uses of ammonia.

2. In a reversible reaction, the forward reaction is when sulphur dioxide reacts with oxygen to form sulphur trioxide and the backward reaction is when sulphur trioxide decomposes to form sulphur dioxide and oxygen
    a) Write the chemical equation on this chemical reaction?
    b) What is the effect of pressure on this chemical equation?

3. Write any three characteristics of equilibrium?

4. Suggest three ways to increase the amount of product in the following chemical reaction.

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} + \text{Heat}$$

5. The first step in the industrial production of the chemical called the king of chemicals is given below. Write the chemical equations for the next three steps.

$$S_{(s)} + O_{2(g)} \rightarrow SO_{2(g)}$$

6. Write the reason for the actions given below
   a) When a little concentrated sulphuric acid is added to glucose, a black substance is formed.
   b) Sulphuric acid is not used as a drying agent in the preparation of ammonia.
   c) While diluting sulphuric acid, the acid has to be added to water in very small quantitues by stirring.

7. Concentrated sulphuric acid is a good oxidising agent. Explain based on the given reactions.
   a) $C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$
   b) $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

8. Find the basicity of the following acids.
   a) Nitric acid
   b) Sulphuric acid
   c) Hydrochloric acid
   d) Oxalic acid
   e) Phosphoric acid
   f) Carbonic acid

9. Caustic Soda is industrially produced by chlor-alkali process.
   a) Write the equation for the reaction taking place at anode?
   b) Write the equation for the reaction taking place at cathode?
   c) Write the equation for the formation of caustic soda?

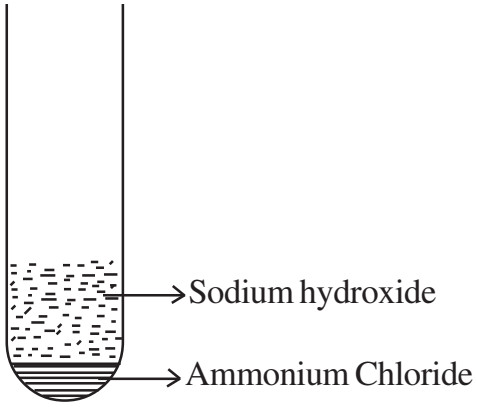
10. a) What is salt hydrolysis?

b) Ammonium chloride is formed by which acid and base?

c) What is the nature of aqueous solution of ammoniam chloride?

IV Short Answer type Question (4 mark)

1.



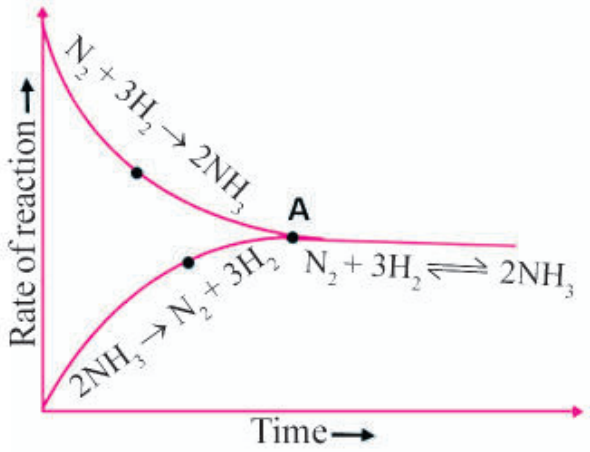
a) What is the nature of gas formed acidic or basic?

b) How this product is industrially prepared?

c) i) What is your observation, when glass rod dipped in HCl is shown in the mouth of the test tube?

ii) Write the Chemical Equation of this reaction?

2. Graph of a reversible reaction is given below.



a) In this Chemical reaction, the point 'A' represents .....

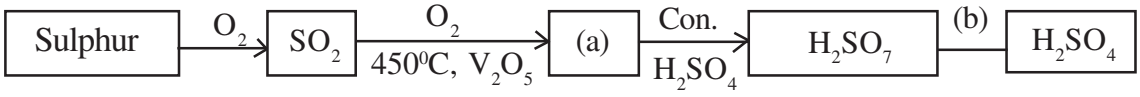
b) With the help of graph, explain, how catalyst influence this reaction?

c) How the following factors influences after reaching 'A'?

i) Pressure increases

ii) NH_3 is removed from the system

3. a) The flow chart of industrial preparation of H_2SO_4 is given below. Identify (a) and (b)?



b) Write the observation of following reactions and identify which property of H_2SO_4 is used here?

Con. H_2SO_4 is added to blue coloured crystals of copper sulphate taken in a watch glass.

c) Write the number of hydrogen ions released during the ionization of a molecule of sulphuric acid?

4. a) Can we use Con. H_2SO_4 as drying agent for the industrial preparation of NH_3? If we used it as drying agent, then write the chemical equation?

b) Write any two uses of sulphuric acid?

c) Some fertilizers are given below.

(Urea, Sodium nitrate, ammonium phosphate, potassium chloride)

Complete the table.

Function in plants	Fertilizers that can be used
To accelerate plant growth	
To increase productivity	
To boost immunity of plants	

Chapter 7

SOME COMPOUNDS OF INDUSTRIAL IMPORTANCE

ANSWER KEY

I Objective type (1 mark)

1. d)
2. b)
3. c)
4. d)
5. b)
6. c)
7. c)
8. b)
9. a)
10. 450^0C
11. c)
12. a)
13. Decreased
14. c)
15. b)
16. c)
17. c)
18. a)
19. d)
20. C)
21. A)
22. b)
23. b)

II Very Short Answer (2 mark)

1. a) NH_3
   b) $2NH_4Cl + Ca(OH)_2 \longrightarrow 2NH_3 + CaCl_2 + 2H_2O$

2. a) CaO
   b) To remove moisture

3. a) Density of NH_3 is lower than air.
   b) Turns Red litmus to blue

4. NH_4OH, $NH_3 + H_2O \longrightarrow NH_4OH$

5. Concentrated NH_3 Solution is liquor ammonia
   Liquified NH_3 is called Liquid Ammonia

6. a) NH_4Cl, $NH_3 + HCl \longrightarrow NH_4Cl$

7. a) $2NH_{3(g)} \rightarrow N_{2(g)} + 3H_{2(g)}$, NH_3
   b) $2HI_{(g)} \rightarrow H_{2(g)} + I_{2(g)}$; HI

8. a) Chemical Equilibrium
   b) Rate of forward and backward reactions are equal, Dynamic in nature

9. a) Sponge iron
   b) Catalyst helps to attain equilibrium very fastly

10. a) Oleum
    b) $SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$

11. It is an exothermic process, so H_2SO_4 initially formed become Smog and it affect the solubility

12. a) Drying property
    b) Dehydrating agent

13. 13.a) A - $NaH\,SO_4$ b) A - $KHSO_4$
B - HCl B - HNO_3

14. a) H_2SO_4 b) 2

15. a) Acid release H^+ ions in aqueous solution and base release OH- ions in aqueous solution.
b) $NH_4OH \longrightarrow NH_4^+ + OH^-$

16. a) Chlor - alkali process
b) Membrane cell

17. a) Na^+, Cl^-
b) Preparing soap and detergent, making paper

18. a) It is an exothermic process, so reactor always be hot
b) Making PVC, Purification of metals

19. a) Acidic - NH_4Cl, $(NH_4)_2\,SO_4$
Basic - Na_2CO_3, CH_3COONa Neutral - KCl, NaCl

20. It should dissolve in water. It should not varry the P^H of soil much

III Short Answer (3 mark)

1. a) $N_{2(g)} + 3H_{2(g)} \overset{Sponge\ Fe}{\rightleftharpoons} 2NH_{3(g)} + heat$
b) At low temperature, the rates of forward and backward reactions become too slow and hence the system will take more to time to attain equilibrium.
c) * Production of chemical fertilizers
* Petroleum refining

2. a) $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$

b) If we increase the P, rate of forward reaction increases

If we decrease the P, rate of backward reaction increases

3. i) Both the reactants and the products coexist at equilibrium
ii) The rates of forward and backward reactions are equal at equilibrium
iii) Chemical equilibrium is a dynamic equilibrium at the molecular level
iv) Equilibrium is possible only in a closed system

4. $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} + \text{Heat}$

$\Rightarrow$ Add more N_2 or H_2

$\Rightarrow$ Remove NH_3

$\Rightarrow$ Increase pressure

5. $\Rightarrow 2SO_{2(g)} + O_{2(g)} \overset{V_2O_5}{\rightleftharpoons} 2SO_{3(g)}$

$\Rightarrow SO_{3(g)} + H_2SO_{4(\ell)} \longrightarrow H_2S_2O_{7(\ell)}$

$\Rightarrow H_2S_2O_{7(\ell)} + H_2O_{(\ell)} \longrightarrow 2H_2SO_{4(\ell)}$

6. a) H_2SO_4 is a dehydrating agent, it absorbs H & O from glucose as H_2O.

b) Because sulphuric acid combines with the base NH_3 and forms ammonium sulphate $[(NH_4)_2\ SO_4]$

c) If diluting is done by adding water to acid, the chance of splashing out of acid leading to accident is very high (Denser than H_2O, Exothermic reaction)

7. a) $\overset{0}{C} + 2\overset{+1}{H_2}\overset{+6\ -2}{S}O_4 \longrightarrow \overset{+4\ -2}{C}O_2 + 2\overset{+4\ -2}{S}O_2 + 2\overset{+1\ -2}{H_2}O$

$\Rightarrow$ Oxidation number of C increased (Oxidation)

$\Rightarrow$ Oxidate number of 'S' in H_2SO_4 decreased (Reduction)

$\Rightarrow$ Here H_2SO_4 act as Oxidising agent

b) $\overset{0}{Cu} + 2\overset{+1}{H_2}\overset{+6\ -2}{S}O_4 \longrightarrow \overset{+2\ +6\ -2}{Cu\ S\ O_4} + \overset{+4\ -2}{S\ O_2} + 2\overset{+1\ -2}{H_2\ O}$

$\Rightarrow$ Oxidation number of Cu increased (Oxidation)

$\Rightarrow$ Oxidation number of 'S' in H_2SO_4 decreased (Reduction)

$\Rightarrow$ Hence H_2SO_4 act as Oxidising agent

8. a) $HNO_3 \rightarrow 1$ b) $H_2SO_4 \rightarrow 2$ c) $HCl \rightarrow 1$

   d) $H_2C_2O_4 \rightarrow 2$ e) $H_3PO_4 \rightarrow 3$ f) $H_2CO_3 \rightarrow 2$

9. a) Anode $\Rightarrow 2Cl^- \rightarrow Cl_2 + 2e^-$

   b) Cathode $\Rightarrow 2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

   c) $Na^+ + OH^- \rightarrow NaOH$

10. a) It is the process by which the anions, cations or both of a salt react with water to form a solution that has acidic or basic nature.

    b) Acid $\Rightarrow$ HCl, Base NH_4OH

    c) Acidic nature (HCl $\rightarrow$ Strong acid, $NH_4OH \rightarrow$ Weak Base)

Short Answer (4 mark)

1. a) Basic

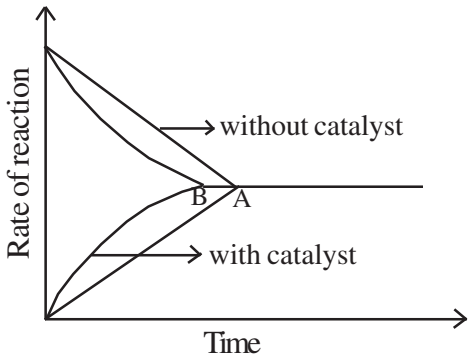
   b) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} + Heat$

   c) i) Dense white fumes

   ii) $NH_3 + HCl \longrightarrow NH_4Cl$

2. a) Chemical Equilibrium

   b)



Catalyst helps to attain equilibrium fastly

c) i) Rate of forward reaction increases

ii) Rate of forward reaction increases

3. a) SO_3 b) H_2O

b) Blue colour, disappears, Dehydrating agent

c) 2

4. a) No. $H_2SO_4 + 2NH_3 \longrightarrow (NH_4)_2\ SO_4$

b) Making fertilizers, Medicines, Cloths, Detergents

c) Urea, Ammonium Phosphate, Potassium Chloride

District Panchayath Kasaragod



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Kasaragod